3rd Applied Active Inference Symposium ~ "Enacting Ecosystems of Shar

~ 1st session | 5:28:56

Hello and welcome everyone.

This is the third Applied Active Inference Symposium

Enacting Ecosystems of Shared Intelligence.

It is August 22, 2023.

We have a lot of really awesome presentations and sessions coming up

in the first interval and in the second interval.

So let's just get right into it with our first talk.

This will be by Andrei Bastos.

So Andrei, thank you for joining and looking forward to your talk.

Thank you very well, Daniel.

And thank you for the other organizers of the Active Inference Conference

and Symposium here.

I'm glad to be talking to you guys today about my work.

It's the title of the presentation is Multilaminar,

Multi-Area Recordings and the Non-Human Primate.

That's these guys right here, the CAC monkeys.

Suggest that predictive coding is implemented via predictive routing.

And so what do we mean by these things?

Well, just as a point of introduction,

I'd like to start by just saying that we started out,

I started out working with Carl Fristen in about 2010.

And we were really interested in trying to map the neurobiology of the predictive coding circuits

that have been theorized and hypothesized onto portico laminar architecture,

which you can see here.

And in that work with him, we proposed that different layers and different oscillations

were involved in either capturing and calculating prediction error or predictions.

And so what we do in my lab, and I recently started as an assistant professor in the Department of Psychology at Vanderbilt.

And so what we do is we train rhesus macaque monkeys on tasks which are more or less predictable

and that have different sensory elements that are more or less predictable.

And we'll get into what that looks like.

And then we use multi-laminar probes, so high-density probes with dozens or hundreds of contacts

that can record across different layers.

And so this is a beautiful drawing here of what the layers look like of cortex

by Santiago Ramon y Cajal over 100 years ago.

And you can see this really beautiful layered architecture.

Layer 1, 2, 3, 4 is this dense one in the middle, 5 you get these large brand little neurons,

and then 6 on the bottom.

And so we're interested in my lab primarily at two levels of explanation.

The first is what do these neurons spike to?

What makes them fire action potentials and excites them?

And do they get more or less excitable depending on how predictable stimuli are?

And second, how do they oscillate?

In which frequency bands do those rhythmic activity tend to cause them to fire more or less action potentials?

And so we record across layers in one area over here, but we might put in our electrodes in several other areas

so that we can also capture aspects of cortical communication.

And these oscillations in this communication we think is really intimately linked to the interaction between inhibitory cells,

these cells in red here, and excitatory neurons, these cells in black that I've written.

Okay, so without further ado, let's just dive right into it here.

So here's the outline of my talk.

In part one, we're going to be talking about first an overview of what do we actually mean by this predictive coding model,

and how do we think it may be mapped onto cortex and paying special attention to the neurophysiology.

So here's the bird's eye overview of predictive coding at the level that maybe a neurobiologist or a cognitive scientist

or a cognitive neuroscientist might need to know about in order to be able to apply it to their own work.

And so here's what I mean by predictive coding, that there's a hypothesized circuit that is especially rich in higher order areas.

So here's like the front of the brain, which learns about the statistical regularities of the world

and creates predictions about what is going to be seen and felt and touched and so on in the next moment in time and in this moment in time,

and sends those predictions to earlier parts of the brain.

So it's a lower order cortex, receive these predictions.

And sensory inputs, the inputs coming from the eyes and ears and all of our sensorium is then compared to this top-down prediction.

And if there's a mismatch, that's a prediction error.

And the prediction error is fed forward.

It's a mismatch response.

And it can be used to drive models, drive these internal representations in order to be better over time and to make better predictions.

So this is the task of working memory, where monkeys are trained to fixate on the center of the screen.

When there's a visual stimulus, in this case, a car stimulus that's shown, there's a brief delay in which that stimulus has to be held in working memory.

When the animal's tested on which of the three items was in working memory, the animal makes a correct response when they cicada, in this case, to the upper left, to indicate that the car stimulus was last shown.

So we then manipulated the predictability of this information that was to enter working memory in the following way.

For 50 trials in a row, we would either keep the sample stimulus constant.

So car, car, car, and that's being depicted by the A here.

Or in another context, there would be an unpredictable sampling regime, where either the car, orange, or green could be sampled at any moment in time.

So the animals went back and forth in being in a predictable context or an unpredictable context.

And then we asked the question, what does brain activity look like when the exact same stimulus is processed in a predictable regime versus an unpredictable regime?

And in order to do this work, we implanted these monkeys with multi-laminar probes.

So these are probes, again, with these, at that time that I did this work, about a dozen or so, or maybe up to 32 contacts.

Nowadays, we can go into the several hundreds of contacts along a single probe.

We can use MRI guidance in order to position those probes perpendicular to cortex, such that they'll span each of the six cortical layers, layer one through six.

Here you can see the white matter here below.

And then we sampled from different parts of both the visual cortex, because remember, those are the areas that might issue prediction errors in the face of an unpredictable sample,

as well as higher order cortex, which has been hypothesized to create these internal models that then issue predictions.

So here's what we found.

We first just analyzed the multi-unit activity, which is a measure of the spike rate.

So how many action potentials per unit of time are these neurons firing?

And what's being shown here is the multi-unit activity as a difference between when that sample was unpredictable minus predictable.

So positive activity means more activity during an unpredictable sample, i.e. a violation of a prediction or a putative prediction error signal.

And all throughout the brain, from V4, the sensory area, all the way to these higher order frontal parietal regions,

we saw without any exception that neurons were much more excitable when they were processing an unpredictable stimulus compared to a predictable stimulus.

And also, not surprisingly, this started already at the feedforward sweep of activity, such that these red bars here indicate the start of significance.

So sure enough, area V4 was the first to show significance, so more spiking to the unpredictable or overpredictable.

And this activity fed forward up the cortical hierarchy.

Next, we looked not only at the firing rate of these neurons, but also the local field potential, which is informative about the local oscillatory activity that's present.

So you can think of these oscillations as massive waves of activity that are synchronous across a large population of neurons,

but of course are produced through the interplay between these recurrent inhibitory bouts of activity.

And so here's what we found. We performed this power spectral analysis, again, comparing power in the unpredictable case versus power in the predictable case.

And we did a percent change metric, very simple percent change metric.

So again, what's shown here on the y-axis is in positive, more power to unpredictable samples, and in the negative values, more power to the predictable samples.

Now, what we saw in the gamma range, so that's everywhere above about 30 or 40 hertz, again, in all areas, very strongly mirrored what we saw in the spiking activity, mainly,

but there was more of this high frequency and also the classic low range frequency in the gamma range.

So in both the low gamma and high gamma ranges, an enhancement of power during the unpredictable cues.

And that was also the case in the theta band, and this makes somewhat sense from the point of view of neurophysiology,

that this theta frequency and this gamma frequency, they very often track one another and do similar things in cortex.

Why exactly that is the case? We actually don't know what the reason for that is, but that's an observation that we've seen in many other tasks.

But what we saw in between the theta and the gamma bands was this other band at between 10 to about 20 or slightly higher in frequency,

this classic alpha-beta band.

And this alpha-beta band acted in the exact opposite way.

Namely, there was more power to predictable samples than to unpredictable samples, with one exception.

Area 7a was an exception. It did have that canonical pattern in the alpha band, but in the low beta band,

it was actually similar to gamma.

And we think that this might be a working memory update mechanism.

But for now, we are really stressing and emphasizing the fact that across all of these areas,

you had this beta signal that was stronger to the predictable case, to the predictable sample.

Okay, so next, we had these laminar electrodes.

So let's look at the difference between what these neurons are doing when you record from the so-called superficial layers of cortex

versus the so-called deep layers of cortex.

So the superficial layers are layers 2, 3.

The deep layers are these layers 5 and 6.

These superficial layers, they are the layers that feed forward.

So remember from that introductory slide that these are the layers that were hypothesized based on predictive coding theory to house the prediction error neurons.

Because according to theory, those prediction error signals are sent in the feed forward direction.

So we then tested that in a second way,

which was to test whether or not those enhanced signals during unpredictable samples are more present in layer 2, 3.

And that's exactly what we found in V4.

So when we again take the spiking signal and subtract unpredictable minus predictable spiking,

we see that the signal is more enhanced in superficial layers of V4.

There was a trend towards that in PFC as well.

But in V4, it was a significant observation.

And furthermore, when we now look at the LFP gamma power,

this power in the 40 to 90 hertz span,

again, we do the power change.

So these positive signals mean there's an enhancement of gamma power to the unpredictable signal.

And again, now in both V4 and PFC,

it's the superficial layers that are having more of this unpredictable signal.

Again, and that's consistent with the theorized type of circuitry for these canonical circuits.

Next, we were interested in whether or not these power changes were blankets across the board

that just overall increase mu activity and gamma activity,

independent of the exact sample that was being shown,

or whether these signals were actually specific to the actual item that was being predicted.

Because remember, in our task design, animals were, for a large portion of time,

predicting that there should be a car stimulus,

or predicting an orange, or predicting the green stimulus.

So in that particular task requirement,

it wouldn't make sense if task representations for all three stimuli

all went up to an unpredictable stimuli.

That would be more consistent with just an overall avousal signal.

But what we were next interested in is whether or not neurons that actually were selected for the car,

if it was those neurons in particular that had the unpredictable signals.

And what we found is that that was indeed the case.

So we performed the same analysis as before,

LFP, gamma power difference for unpredictable minus predictable,

but now as a function of whether or not those neurons preferred the stimulus that was being predicted.

And what we found is that the preference actually modulated this gamma band activity very strongly.

And so it was this unpredictable signal change was more specific to neurons

that actually preferred that predicted stimulus,

indicating that this prediction was not acting as a blanket,

similar to an arousal mechanism,
but was actually reaching down at the level of the representations
that were specific to the thing that needed to be predicted.
And that was even more the case in the beta frequency range, actually,
where the neurons that didn't encode the predicted stimulus,
they couldn't care less whether or not that stimulus was predictable.
It was really only neurons that carried information about the stimulus that was being predicted.
Those were the neurons that were not enhancing,
but diminishing their beta during an unpredictable signal,
or said another way, enhancing their beta to a predicted signal change.
So here's what we think is going on.
We call this neurophysiological implementation that we found of predictive coding,
we call it predictive routing, to make a semantic point,
but also a slightly more neurophysiological and computational argument
that we don't actually find strong evidence for this mismatch and prediction signal at the level of neurons.
But instead, we think that this prediction circuit exists,
but uses these oscillations in order to implement what is predicted.
So here's how we think this works.
So when a prediction matches a stimulus, as is being shown here,
when you're predicting A and you get a stimulus A,
then the A column, in other words, it doesn't have to be a column,
but the area of the brain or the neurons that encode A
are going to become prepared to receive the stimulus A
by enhancing the power of this alpha-beta signal.
So this top-down beta from higher order areas, we think,
prepares a cortical circuit to receive a predicted stimulus.
And it prepares that stimulus by enhancing, that's this up arrow here,
the alpha-beta, especially in the deep layers,
which can inhibit both the gamma and the spiking signal in the superficial layers.
As a result, so this top-down signal inhibits the superficial layers.
And as a result, there is decreased gamma and spiking
and decreased feed-forward output in the case of a predictable stimulus.
What is the case of an unpredictable stimulus?
Well, stimulus A, just like before, is still going to be coming in.
But because it's not predictable, that column,
that set of neurons that normally represent A,
was not previously prepared by an alpha-beta signal.
And so the alpha-beta signal is relatively low in an unpredictable context,

releasing the inhibition that the deep layers exert on superficial layers.
And as a result, causing more gamma, more spiking, and more feed-forward output.
In other words, what we are hypothesizing here
is that predictive coding and this error computation
comes about as a failure in predictive inhibition.
And as a result of the lack of these mechanisms being in place
during an unpredictable stimulus,
there is an enhanced gamma output.
And that's what we call a prediction error,
or that's what we interpret as a prediction error.
So here's the predictive routing model one more time.
In a predictable stimulus, top-down inputs prepare the column to receive a stimulus.
That preparation leads to inhibition, less feed-forward output.
In an unpredictable case, those mechanisms aren't in place.
As a result, more feed-forward gamma output.
So that's part one.
That's our cortical neurophysiological hypothesis then
that we call predictive routing
of how we believe predictive coding is implemented in the brain.
And next, I want to turn to a little bit more
about how these rhythms are actually produced in the brain.
Because one of the things that we noticed
from just placing our laminar electrodes into cortex
is that independent of where we put our probe,
so here's an example of prefrontal cortex activity,
we would see these high-frequency gamma waves in superficial layers
and slower-frequency alpha-beta waves in these deep layers.
And you can do this analysis very easily and quickly
to reveal this, what we call, spectral laminar pattern.
And what this is, is simply dividing each contact on the probe,
dividing the power that you observe at that contact
by the contact with the maximum power along the probe.
And what this does is it normalizes power
such that a value of 1 is the layer
that had the most power at that particular frequency band.
And so if you look here on the x-axis, this is frequency,
so these red colors are tracking which particular layer
has more power at that frequency.

So you see more of this low and high gamma power
in the superficial layers, so that's above 0.0 on the y-axis,
and more of this alpha-beta activity
in the deeper layers of the cortex.
And so in a large-scale survey of cortex,
together with Bob Desimone's lab and Diego,
and so I did this partially while I was at MIT
when I was a postdoc with Earl Miller,
and I continued this now at Vanderbilt.
So this was a really cross-institutional study
and involved multiple labs.
And the reason that that was important
is that most labs specialize
in being a prefrontal cortex lab
or a V4 lab or a LIP lab, et cetera.
And relatively few investigators
had actually done the necessary work
to be able to record with these laminar probes
in multiple areas simultaneously.
And so therefore, by coordinating
and pooling data for multiple experiments,
we were able to record from a total of 14 areas
in the study that we have now in review.
And what we found was that independent of the area,
we found a spectral laminar pattern
that was highly conserved across areas.
And so here it is from 0 to 150 hertz
for area V4 in the sensory cortex,
7a in parietal cortex, prefrontal cortex
in the front of the brain.
Everywhere we looked, we found
that there was this enhanced gamma power
in these red lines here in superficial layers
and enhanced alpha-beta power in the deeper layers.
And what we then turn to next is,
first of all, well, let's really confirm
at a histological anatomical level
that it's really these layers,

that it's, is it layer one?
Is it layer two?
Is it four that expresses gamma?
And the same question for beta.
Let's really confirm that that is in fact the case
because it seemed like a very prominent hypothesis
and a very prominent observation.
We wanted to really confirm it.
So that's the first part of this, part two.
And then we're going to get into
which specific mechanisms
might generate these oscillations
and how does that inform computational modeling.
And so then this was work done in my lab at Vanderbilt
led by a very talented undergraduate student, actually,
Max Lichtenfeld.
And so we performed these histological reconstructions.
Here's the probe, an example probe,
where we perform these tiny electrolytic lesions
in particular areas while we were recording.
Here's the overlay of the laminar boundaries.
So in red here, this is layer four.
And then, of course, we had the electrophysiology.
So we could look at the gamma power,
gamma relative power, the beta relative power,
and the point at which they cross over,
actually we determined,
was pretty consistent in layer four.
And of course, then we also mapped on
the other features of these things to the layers.
And so here's what we found,
is that the crossover,
the point at which the gamma profile in red
and the beta profile in blue,
the point at which they cross over,
at which their powers are equal,
that tended to land in layer four.
The peak gamma power tended to land

in layer two or three,
and the peak beta power tended to land
in layer five or six.
And so we mapped these distances
between these different electrophysiological signatures
or distinct profiles to the different layers.
And we did this for several different probes
and for several different animals
until we were able to get
a very nice distribution here,
which confirmed that indeed,
the gamma beta crossover is consistent
with being at laminar position zero,
we call the middle of layer four.
Whereas this gamma is at the boundary
between layer two, three,
and beta at the boundary
between layer five and six.
And so here is our current toy model
of what we think is going on.
So we think that there's different inhibitory neurons
that are interacting
with their excitatory cell partners
to generate fast frequency gamma activity
in the superficial layers.
And in turn, that there is enhanced,
that there's another type of interneuron
that is generating with slower time constants,
this alpha beta rhythmic activity.
So in order to examine this,
we again turned to anatomy.
We again turned to this question,
well, then which exact inhibitory neurons
are most prominent, therefore,
in the layers that express
these different frequencies?
And in order to perform this work,
what we did is we quantified

the laminar expression
of three distinct interneuron subtypes
that are known to account
for over 90% of the inhibitory interneurons
in the primate brain.
And those inhibitory neuron subtypes
are parvalbumin positive,
calretinin calbindin positive.
And importantly,
previous computational modeling work
had really singled out
the parvalbumin cell class
as the critical class
for generating these gamma oscillations
because of their fast time constants
and strong synapses
onto the cell bodies
of excitatory neurons.
And so again,
we performed this painstaking
histological analysis
where you perform different stains,
mounting, cover slipping,
microscopy,
semi-automated thresholding,
and cell counting
in order to be able to then
really have a quantitative view
on which layers have,
which cell types
that are most associated
with these frequencies.
And so here's what this data
then looks like.
We have the gamma
and alpha-beta electrophysiology here.
And then at the same time,
we have the neural,

the presence of different neuron types
with the three inhibitory cell classes here.
And then we can perform an analysis
to determine
which of these cell classes
is most strongly associated with the gamma.
And here's what we found.
That in both the low gamma
and the high gamma
across over 14 regions,
what we found quite significantly
was that the,
it was really the parvalbumin cell counts
shown here
as a function of laminar depth
that were giving us
the best explanation
for the laminar profile
of the gamma rhythmic activity,
both when you take a very wide band
as well as when you perform
a low versus high gamma split.
It comes out in both ways.
So how can this inform
computational models of oscillations?
Well, here's what we think
is going on then.
We collaborated with several people
here on this work,
including Sanchez Toto et al
and computational modelers in Spain.
And what we did was
we performed an analysis
where we took
different interneurons
with different types of connectivity.
So these PV,
these parvalbumin interneurons

were modeled as
having strong
inhibitory connections
to the pyramidal cell class
and strong
self-inhibition,
which has been important
for gamma oscillations previously.
And when you hook them up
in these particular ways,
sure enough,
this,
and you measure
the spectral output
of these neurons,
these P2 neurons,
superficial
pyramidal neurons,
sure enough,
they express gamma.
And when you put in
different types
of interneurons
with slower
time constants
and different types
of connections
to these deep layer neurons,
sure enough,
they come out
with more
alpha-beta oscillations.
So indeed,
this anatomical work
could produce
even more,
in the future,
even more

accurate models
of how these interneurons
work together
and with pyramidal cells
across different layers
in order to
express these oscillations.
So this is really
a work in progress
and one of the first
steps towards this.
And our ultimate aim
is to really pull together
this rich anatomical
and neurophysiological data
together with
computational models
to be able
to approximate
the laminar structure
of cortex
to a better
and better degree.
So this next part here,
I'll talk a little bit
about a causal test
now of the
predictive routing model
with propofol anesthesia.
So this will be
the last part
of the talk here.
And so we were
interested in anesthesia.
I got interested
in anesthesia,
actually,
through my work

with and contact
with Jake Donahue
over here,
who is a very talented
MD-PhD student
at MIT
at the time
and was being
co-mentored
by Emery Brown
and Earl Miller,
who was also my mentor.
And so we became
interested in this topic
for the following reason.
Well, first of all,
as surprising
as this may seem,
if you go into
a surgical clinic
and you're administered
an anesthetic
such as propofol,
well, it turns out
that the anesthesiologist
may not know
exactly the cortical mechanisms
and the systems level
mechanisms for how
that anesthetic
is being,
is actually working
in the brain.
And it's actually,
to this day,
remains
more often than not
the case

that there's actually
no intraoperative
monitoring of brain state.
It's far and few
between the anesthesiologists
that are actually
doing that.
And so Emery Brown,
who's an anesthesiologist
himself,
is really pushing
for more and more
adoption of simple tools
like EEG
and spectral estimation
methods
by the clinic.
And so our work
was clinically motivated,
but it was also motivated
by the ability
to test this predictive
routing model
when activity,
especially in prefrontal
cortex areas,
is relatively silenced.
And so in these experiments,
we administered propofol
to, again,
these rhesus macaques
and we recorded brain activity.
There's these local field potentials
and spiking activity
in the awake state
as well as
in the unconscious state.
And what we first noticed

is that in the unconscious state,
there were these
slow rhythmic bouts
of activity
at about one hertz,
one cycle per second
or slightly
little faster,
a little slower than that.
Those are the slow frequencies.
And these slow frequencies
organize these massive bouts
of spiking,
which could sometimes
be very synchronous
across cortex.
but that there were
long periods
of silence
in between
these bouts
of spiking
in the unconscious state.
And so
propofol anesthesia,
we found,
is accompanied by,
again,
these strong increases
in slow frequency
activity,
which create alternating
on and off periods
for spiking.
There's reduced spiking
activity
throughout the network,
but the strongest reductions

are in prefrontal cortex.
And sensory areas
are relatively spared.
So they're still
quite reduced
in their excitability
in the unconscious state,
but prefrontal cortex
is more inhibited.
There was also,
we found,
reductions in cross-area
synchrony
and very pronounced
reductions
in the alpha-beta band.
And so as a result,
we thought,
hmm,
we could actually
use this model,
this propofol anesthetic model,
in order to test
certain aspects
of predictive routing.
Because what's going on here,
we have lower order cortex
that's hypothesized
to feed forward
prediction errors
to higher order cortex.
And in turn,
these higher order areas
might feed back
their predictions.
But as a result
of propofol,
these higher order areas

are profoundly impaired
and very reduced
in their spiking activity.
And the alpha-beta rhythm
is also the synchrony
between areas
is highly reduced.
So what's going to happen
in this case?
Well,
if you think
from the point of view
of simply everything
is going down,
then sensory areas
in a response
to an unpredictable stimulus
should also just go down.
But if you think
from a predictive coding
or a predictive routing
point of view,
the function
of these higher order areas
is to create feedback
predictions
that inhibit activity.
So in the absence
of those feedback
inhibitory activities,
you might actually
hypothesize
that there would be
an enhancement
of the feedforward response
when that feedforward response
is characterized
by an unpredictable stimulus

or prediction error.
And so we sought
to test this out
using the simple sensory
prediction paradigm
where,
whoops,
we have a tone stimuli.
These tone stimuli
could either be repeated
or occasionally
there could be
an oddball stimulus
or beep, beep, beep, beep, beep, beep, beep, boop.
And what we would expect
is that there would be
enhanced feedback predictions
when stimuli
were repeatable
and predicted
and that the occasion
of an oddball
would signal
an enhanced gamma
feedforward signal.
So we performed
the auditory stimulation task
both in the awake state
as well as after
there was an administration
of propofol
leading to loss
of consciousness.
And here's what we found
in the local state
just as we found
in the visual cortex
for an unpredictable

visual stimulus
as in the visual cortex
so in the sensory cortex.
My PhD student,
Sophie Shang,
!
who performed this analysis
found an enhanced gamma.
So right at the time
of the local oddball
so we have time here
on the x-axis
frequency on the y-axis
in the spectrogram
hot colors mean
more activity
to an unpredictable
stimulus
so there was an enhanced
gamma response
to this unpredictable stimulus
decreased alpha-beta
to the unpredictable stimulus
or said another way
when the stimulus
was predicted
there was more alpha-beta
and here's what happened
in the anesthetized state
when we performed
this exact same contrast
we lost the predictable
alpha-beta signal
and instead
everything was enhanced
so there was a decrease
in frequency
towards the low gamma range

so the cortex
just essentially went
into a constant ringing
in the gamma frequency range
which to our surprise
was much stronger in power
compared to the gamma
that we saw being enhanced
during an unpredictable
stimulus in the awake state
and the way we interpret this
is that in the absence
of this inhibitory
predictive alpha-beta signal
this gamma signal
becomes uncontrolled
and enhances itself
in the local anesthetized state
here in area
TPT is a sensory area
I should have mentioned that
so now to conclude
we propose predictive routing
so predictive routing
is a neurophysiologically inspired
implementation of predictive coding
which stresses the importance
of these alpha-beta signals
for implementing predictions
and inhibiting sensory areas
and areas of the brain
that are experiencing
a predictable stimulus
in the absence of that
you get an enhanced
feed-forward gamma output
and this distinction
between gamma and beta

across layers
we found was ubiquitous
across cortex
really suggesting
that something like this
may in fact be
a truly a canonical
circuit motif
that's repeated
across the brain
and these alpha-beta rhythmic
predictive preparation
or inhibition
we found
we observed
was lost
in the unconscious state
with unconsciousness
being mediated
by propofol
which by the way
is one of the most
often used anesthetics
and that as a result
of this loss
of predictive
predictive alpha-beta
there was an enhanced
gamma modulation
to these unpredictable inputs
in other words
the sensory cortex
was in a
in a disinhibited state
and so with that
I'd like to thank
my many collaborators
and the Bostos lab

at Vanderbilt
as well as collaborators
on the anatomy
and collaborators
at MIT
as well as my
financial support
and lastly
we just started the lab
about two years ago
we're still in
recruiting stage
so you guys can check out
my website
my x slash twitter feed
and email me
if interested
in contributing
to this line of research
and so with that
I'll turn it over
in case there are any questions
thank you for your attention
awesome
thank you Andre
I'll just ask one
quick question from the chat
Dave Douglas asks
does propofol
change the activity
of neuronal microtubules
in any interesting way
thanks Andre
I'll just ask one
question from the chat
Dave Douglas asks
does propofol
change the activity

of neuronal microtubes
in any interesting way
I haven't looked into
the microtube
kind of hypothesis
that's one of the hypotheses
advocated for
for many years
by people like
Stuart Hammerhoff
and there has been
some recent work
done by
by George
Meshure
I believe
and
so you should go
check out
some of his papers
on that topic
but I haven't
looked into that
and
I am
I remain somewhat
skeptical
to the
quantal
explanation
for
consciousness
but
mostly because
the
mechanism
doesn't appear
at first glance

to be
very biologically
reasonable
and
and also
that there's not
a lot of
there's essentially
not very much
data on it
but that may change
in
in the coming years
so
something to look out for
great
one
one other question
more generally
how do you see
experimental work
in the active
inference ecosystem
and how do you see
the synthesis
of
theoretical
and in-principle
work
with this kind
of real
wet lab
and real
recordings
on the neurons
type analysis
that you're doing here
yeah

so
it's a great
it's a great question
I think that
what we need to be doing
as a community
is
having more interactions
between
theorists
and experimentalists
and so
we wouldn't have
performed
for example
this experiment here
if we hadn't
already been
inspired by
biological
and theoretical
models
of predictive
coding
and
and of active
inference as well
and so
that's really
what we need
and
is to
make
not only
hypotheses
that are
turned into
great

theoretical
and computational
infrastructures
and
models
but we also
need to be
thinking about
if we care
about the
biology
and we care
about
trying to
figure out
the brain
then we should
also be
trying to
map those
things onto
neurobiology
and psychology
right
and so
that
involves
them
both
considering
this
neurobiological
elements
that may
implement
and mediate
active inference
in the brain

as well
as the
psychological
and cognitive
constructs
that we think
are going
to be
particularly
sensitive
to capture
certain elements
of the
theory
and so
I remain
very
very welcoming
to that type
of suggestion
and
in fact
there's going
to be a
postdoc
joining the
lab soon
Eli Sinesh
who has been
working
very deeply
in the
active inference
community
and so
we'll have
an opportunity
there to

both work
on theory
guided
data analysis
but also
perform more
theoretically
guided
experimental
designs
that can
go
further
into
the
neurobiological
implementation
and so
this was
here what I
presented
really simply
the first
and small
somewhat
step
towards
linking
the
neurobiological
implementation
level
with the
algorithmic
and computational
level
of predictive
coding

and active
inference
and so
the more
interactions
that we can
have
between
the
communities
the better
that that
will
be
I think
what would
be
somewhat
of a shame
is if the
active
inference
community
went way
too far
into just
pure
theory
and that
and delved
into
increasingly
increasing
areas that
were divorced
from the
biology
so that's

of course
one of
the great
critiques
of the
deep
learning
and more
purely
machine
learning
driven
approaches
is that
they took
a little
bit of
biological
inspiration
but then
started using
algorithms
like
backprop
that were
very
biologically
unrealistic
and so
of course
active
inference
is a
response
to that
in certain
ways
but in

order to
continue
being that
I think
there needs
to be
indeed
a tight
dialogue
between
experimentalists
and theorists
and so if
you have
any particular
ideas
I'd be
happy to
hear them
please
reach out
to me
by email
or
through
X
which you
can find
here
on the
slide
that's
that's
really
cool
that's
awesome
you're

going to
be
working
with
Eli
and one
final
question
this is
from
Alexandra
is it
possible
to detect
the
contribution
of
glia
to the
neural
firing
rate
of the
field
potentials
that is
a question
that I
have not
considered
yet
so we
normally
do not
consider
glia
to be
active

in the
network
but of
course
that
that is
likely to
be an
oversimplification
we know
that they
can have
interactions
with neurons
we know
that they
can also
have
different
levels
of
excitability
and of
course
that's
how they
can
regulate
neurons
metabolically
but as
far as
the
contributions
of glia
directly
to the
local

field
potential
I am
not aware
of how
to
determine
whether or
not they
make a
significant
contribution
the
I
actually
haven't
seen
any
work
on that
but
thank you
for the
question
it's
always
good
to
reconsider
our
core
assumptions
about
the
cell
types
that
are

most
important
to
capture
so
it's
certainly
something
to think
about
thank you
thank you
all right
thank you
Andre
super
cool
work
and good
luck
with
everything
till
next
time
bye
bye
bye
all right
next up
is going
to be
Keith
Dugger
who has
submitted
a pre-recorded
video

the pre-recorded
video is
called
active
inference
and the
actor
model
so I
will bring
that up
right now
okay
here
comes
Keith's
pre-recorded
video on
active
inference
and the
actor
model
active
inference
and the
actor
model
hello
hello
I'm
Dr.
Keith
Dugger
platform
CTO
at
X-Ray

an
extended
reality
AI
company
and
co-host
of
Machine Learning
Street
Talk
podcast
now
as we
all work
to build
an
ecosystem
based
on
active
inference
software
will
obviously
play
a
foundational
role
to
make
the
most
of
active
inference
we'll
need

to
use
software
engineering
paradigms
that align
with the
principles
of active
inference
and I
think there
is one
tailor-made
for our
needs
it's called
the
actor
model
active
inference
and the
actor
model
are two
deeply
connected
understandings
of the
world
they provide
foundational
frameworks
for dealing
with the
dynamics
of complex

systems
with a
focus on
autonomous
agents
that interact
in an
ecology
of nested
systems
I'd like
to
afford some
of their
key
connections
including
the role
of agents
concurrency
autonomy
uncertainty
and behavior
adaptation
we will
see that
active
inference
and the
actor
model
are both
paradigm
shifts
away
from a
deterministic
centralized

step-by-step
thinking
to a
decentralized
networked
concurrent
perspective
of both
computation
and cognition
just a little
bit of history
about the
actor model
back in
1973
Carl
Hewitt
Peter
Bishop
and Richard
Steyer
were all
working at
the Massachusetts
Institute of
Technology
AI lab
to
fundamentalize
a concept
of concurrent
computation
that included
both structure
as well as
adaptable
algorithm

execution
conventional
methods at the
time lacked
robustness
and secure
mathematical
foundation
their collaborative
effort ultimately
led to the
creation of
the actor
model
at the
time
it was
viewed as
revolutionary
due to its
characteristics
of both
increased
error tolerance
and distributed
computation
abilities
throughout the
1980s
and the
1990s
the actor
model became
the basis
for numerous
research projects
as well as
practical projects

gaining popularity
for its
flexibility
and intuitive
approach to
concurrent
computation
computation
it was
primarily
used
from
artificial
intelligence
and
multi-agent
systems
sound familiar
new actor
based languages
like actor
salsa
and early
contributed to
the refinement
of the model
shaping it into
a more robust
and flexible
approach to
concurrent computing
and it remains
alive and well
in computer science
today
more recently
the actor
model is

getting renewed
interest
mainly due to
the growing
need for
distributed
systems
cloud computing
and edge
computing
fueling the
internet of
things
and web
3.0
these computer
tasks are
ideally suited
to take advantage
of the actor
model's
architecture
which is
designed
exactly
for modeling
concurrent
handling
of both
large volumes
of data
on the one
hand
and fine-grained
disparate
autonomous
systems
on the other

hand
this application
of the actor
model has had
profound effects
on major
companies that
utilize its
principles
to handle
big data
problems
such as
twitter
facebook
linkedin
so what
does this
have to do
with active
inference
i'm guessing
you've already
heard some
of the parallels
in the intro
well let's
start by
looking at
a few
core principles
of the actor
model
and how they
relate to the
principles of
active inference
let's start with

the concept of
isolation
isolation means
that an actor
in the actor
model does not
share its state
with any other
actor
it can only be
affected by
receiving a
message
and it can only
affect change
in the state
of other actors
by sending a
finite number
of messages
in response
from a
software engineering
point of view
this isolation
principle limits
potential side
effects of an
operation to a
single actor
thus improving
the system's
overall predictability
reliability
and most
importantly
if embraced
fully

can actually
simplify the
design
looking at
the diagram
we see an
ecosystem of
actors sending
messages to a
particular actor
which in turn
sends back
messages to
other actors
where is active
inference
well
let's recast
receiving and
sending messages
to a perception
action cycle
and denote
external
internal
sensory
and active
states
inference
and we now
clearly have
the necessary
foundation of
active
inference
a markup
blanket
the actors

of the actor
model
amount directly
to the agents
of active
inference
in addition
the finiteness
the fact that
an actor can only
send a finite
number of messages
in response
is also an
important
share of
property
because active
inference models
reality
it necessarily
respects the
resource constraints
of real systems
and this is
nicely bathed
into the
foundation
of the actor
model
let's look at
another core
principle
asynchronous
message passing
communication
between actors
is asynchronous

this means
an actor
doesn't wait
for a response
after sending
a message
it continues
working
it continues
living
as it
were
this
is critical
as it
decouples
the actors
leading to
a system
that can
continue
functioning
living
and making
progress
even when
parts of the
system
are slow
or even
temporarily
unavailable
professor
friston
has said
that the
free energy
principle

is the
ultimate
existential
question
if things
exist
what must
they do
well the
actor model
claims
they must
not wait
on others
of course
an actor
can choose
to wait
on others
but it
must not
be forced
to do so
in the
model
it must
be free
to choose
this leads
us to
another
critical
principle
that both
models
share
autonomy
the free

energy
principle
is a
model
of
physical
reality
and our
reality
is
after all
concurrent
all
throughout
infinite
space
systems
are
evolving
simultaneously
according
to their
local
dynamics
and
therefore
this is
reflected
at the
heart
of the
free
energy
principle
of course
a model
of computation
need not

constrain
itself
to
physics
but
hewitt
et al
were
seeking
to
develop
a model
that
did
model
the
reality
of
distributed
concurrent
systems
and luckily
for us
the actor
model
embraces
both
concurrency
as seen
from the
principle
of isolation
and actor
autonomy
making it
compatible
with active
inference

next
we come
to
nesting
the actor
model
allows
for an
actor
to
not only
receive
and send
a finite
number
of messages
to perceive
and act
it also
allows
as an
action
the creation
of a finite
number
of new
actors
these actors
can either
be nested
within the
parent
say the parts
of an animal
cell
or it
can be
released

into the
environment
as independent
actors
from then
on
this
principle
of the
model
fits nicely
with the
beautiful
concepts
of
multi-scale
nesting
and active
inference
this allows
for actors
to contain
ecosystems
of actors
both all
the way
down
and all
the way
up
last
I want
to cover
two more
actor model
design
principles
behavior

change
and persistence
actors
have the
ability
to change
their behavior
in response
to a message
this adaptability
allows
for the
construction
of complex
stateful
entities
that can
evolve
over time
and indeed
it allows
for entire
ecosystems
to evolve
new
emergent
behaviors
when used
for software
engineering
this adds
a powerful
tool
for managing
complex
dynamic
systems
active

inference
of course
embraces
this to
the extreme
the very
essence
of thing
this is
the ongoing
attempt
to predict
an adapt
to an
environment
and thereby
continue to
exist
to maintain
one's
markup
blanket
in a
broiling
sea
of activity
along
with this
also comes
the concept
of
persistence
persistence
allows actors
to save
their state
and to
restore

or modify
it later
a feature
that embodies
the principle
of memory
memory
is a
prerequisite
for learning
and adaptation
an agent's
ability to
predict
depends on
its ability
to remember
past experiences
and thus
minimize the
surprise
associated with
unexpectedness
the vital role
of memory
is also
emphasized
when we
assume that
agents have
inductive
priors
either from
experience
or inheritance
contributing to
their world
model

this world
model both
guides their
current behavior
and it is
continuously
updated
based on
new experiences
contributing to
their ongoing
adaptation
and existence
okay
great
you say
there are
clear and
deep
connections
between the
actor model
and active
inference
but
how does
this help
us
in the
active
inference
community
well
firstly
in my
opinion
it is a
software

engineering
paradigm
we should
embrace
and if we
do
there are
of course
actor model
libraries
and frameworks
that we
can use
such as
ACA
Orleans
Thespian
Actix
Proto-Actor
and many
more
which we
can immediately
use when
building
active
inference
software
modules
and
applications
there are
also
libraries
languages
and
language
features

that align
very well
with the
actor model
principles
such as
0 and
Q
Tokyo
and Rust
Erlang
Async
Await
and C
Sharp
etc

!
This is
evident
this is
evident
not only
from the
alignment
of the
core
principles
we
previously
discussed
but also
from the
insights
that
active
inference
and the

actor model
bring to
each other
for
example
consider
what is
now
called
Hewitt's
Law
informally
stated as
everything
is
everywhere
this
law
signifies
the
idea
that
in a
truly
asynchronous
distributed
system
it can
take an
arbitrary
amount
of time
for a
message
to go
from
one
place

to
another
and
any
actor
must be
prepared
for that
event
there is
simply
no such
thing
as
instantaneous
in such
a
system
and
no
component
can
make an
assumption
about the
timing
of another
component's
actions
in fact
one must
act as
if a
message
may
never
arrive
this

has
important
implications
it's
just
that
it is
impossible
to
accurately
and
consistently
determine
the state
of the
entire
system
at any
given
time
because
the
information
just
may
not
have
even
propagated
across
the
system
and
also
attempts
to
implement
global

synchronization
will
inevitably
introduce
bottlenecks
and reduce
efficiency
Hewitt's
law
emphasizes
the need
for systems
to be
designed
in a
way
that they
can
effectively
handle
these
unavoidable
delays
and
uncertainties
highlighting
the
importance
of
robust
non-blocking
communication
mechanisms
and local
decision-making
abilities
in short
effector model

systems
are by nature
non-deterministic
does this
sound familiar
what
other
paradigm
highlights
operation
under
uncertainty
and the
autonomy
to continue
despite
the
environment
active
inference
and the
free
energy
principle
active
inference
reflects
the
reality
of an
unpredictable
world
in which
our
software
systems
operate
different

outcomes
may result
from the
same
initial
conditions
due to
the
occurrence
of
events
in a
random
unpredictable
order
this
is the
concept
of
surprise
that we
all know
well
where an
agent
updates
its
beliefs
about
the
world
when the
sensory
input
it receives
does not
match
its

predictions
both
the
actor
model
and
active
inference
acknowledge
that the
world
is
unpredictable
even
more than
acknowledge
it
the
models
accept
this
uncertainty
as a
given
and not
something
to be
managed
away
indeed
as we
know
in the
free
energy
principle
the
uncertainty

that we
maintain
in our
models
is what
gives us
the
flexibility
to
adapt
perhaps
this is
just my
personal
flight of
fancy
what I
imagine
a future
where
software
modules
guided by
active
inference
do
away
with
hard
coded
error
handling
and
swap
in
probabilistic
learning
algorithms

that
optimize
themselves
as the
error
landscape
evolves
modules
that are
robust
and
self
healing
distributed
systems
with no
single
points of
failure
that focus
on
predictive
disaster
avoidance
rather
than
reactive
disaster
recovery
looking
towards the
future
we as a
community
have the
potential to
push the
boundary

of both
active
inference
theory
and
practical
implementations
of the
actor
model
by leveraging
the strengths
of these
two paradigms
together
we can
create software
systems that
are robust
adaptive
and more
aligned
to the
physical
world
in which
they actually
operate
imagine the
future where
software
components
using
active
inference
in the
actor
model

can
anticipate
potential
issues
learn
from past
mistakes
and adapt
in real
time to
environmental
changes
with this
approach
we can build
systems that
are fundamentally
more resilient
and more
efficient
in my
opinion
this can
bring a
step
change
in software
reliability
and performance
and scalability
and heralds
a new era
of computing
weaving
principles of
biology
and cognition
into the

fabric
of our
software
systems
bringing them
closer to
life
in the
process
in conclusion
the coupling
of active
inference
to the
actor
model
provides
a
powerful
new lens
through which
we can
look at
software
design
and
engineering
whether we
leverage
existing
languages
and libraries
aligned with
active
inference
or invent
new ones
we are

standing on
the break
of an
exciting
frontier
so let's
seize the
day
have a look
at the
actor model
and its
relationship
to active
inference
and let's
shape the
future of
intelligent
distributed
computing
thank you
for listening
awesome
great talk
by Keith
thank you
Keith for
sending that
in
there were
some
comments
in the
chat
so Keith
potentially if
you want to

join for a
Q&A
some future
time
but really
cool
presentation
all right
the next
presentation
is by
Sanjeev
Namjoshi
and
this
presentation
is going
to be
called
developing
next-gen
active
inference
tools
broadening
accessibility
educational
resources
and the
software
ecosystem
I'm going
to start
this talk
right now
hello
everyone
and thank

you for
being here
my name
is
Sanjeev
Namjoshi
I'm just
going to
restart it
I'm a
machine learning
engineer
working at
the AI
services
for
just because
it's a
little
quiet
all right
restarting
the talk
hello
everyone
and thank
you for
being here
my name
is
Sanjeev
Namjoshi
and I'm
a machine
learning
engineer
working at
the AI

services
firm
Kungfu
AI
where I
primarily
focus on
computer
vision
projects
today
I'm going
to be
talking to
you
about
my
progress
toward
providing
greater
accessibility
visibility
and
knowledge
of active
inference
and the
free
energy
principle
I'm
excited to
be
presenting
this
material
at the

active
inference
symposium
this year
because the
idea of an
enacted
ecosystem
of shared
intelligence
perfectly
captures
the philosophy
that underlies
the projects
I'm currently
engaged with
I've spent
the last
seven months
on sabbatical
for my job
to work
exclusively
on an
active
inference
textbook
and related
tools
presenting
chapter
presentations
code
reviews
and receiving
feedback
weekly

at the
active
inference
institute
the
institute
has
provided
a
space
where
interdisciplinary
research
can
flourish
as the
connections
and influences
of active
inference
spread to
other
fields
it has
consistently
fostered the
spirit of
collaboration
and shared
intelligence
that I wish
to embody
in my own
work as part
of this
ecosystem
I intend
to continue

closely working
with the
institute
to provide
materials
that will
bring active
inference
to a much
wider
readership
I originally
chose this
project
when I saw
the great
potential in
the active
inference
field
and couldn't
help but make
a comparison
to the state
of deep
learning in
2006
when neural
networks were
one of just
many possible
models
rather than
the dominant
choice in
academia
and in
industry

this of course
all changed
in 2006
when Hinton
and his
colleagues
released the
deep belief
network paper
which is generally
understood as a
start of deep
learning as we
understand it
today
after some
hardware
innovations
and the
release of
the well-known
ImageNet
library
we started to
see coverage
and success
of AI
in the news
as well as
in academic
research
but in 2012
deep learning
truly provided
its value
with AlexNet
and for the
first time

deep learning
achieved better
than human
performance
on image
detection
tasks
but this was
just a start
what followed
was a
proliferation
of deep
learning
all across
industry
and research
I've added
in some
well-known
milestones
just to
highlight
the explosion
of progress
in deep
learning
in the last
decade
though there
was so much
more here
that we
could discuss
so what
about the
state of
active

inference
as a
field
from my
perspective
active
inference
lies in
the same
position
as deep
learning
in 2006
influential
and on
the brink
of exploding
in popularity
this paper
which is
from
contributors
at the
active
inference
institute
shows the
current
growth
of
publications
in the
institute
and its
community
in active
inference
in the

last three
years
the active
inference
field
has seen
a number
of important
milestones
here I
show just a
few that
broaden the
scope and
attention to
the field
we had the
first international
workshop on
active inference
in 2020
we had the
first active
inference
symposium
and the
founding of
the active
inference
institute
then called
the active
inference
lab
we had the
release of
the par
et al

2022
textbook
and the
pymdp
python package
and I
see myself
now as
perfectly
poised to
bring active
inference to
greater visibility
and attention
this is in
part because of
the current
academic interest
in deep
reinforcement
learning and
generative
modeling
working alongside
the institution
and other
organizations
my aim here is
to provide some
of the fundamental
materials to
capture the
attention and
notice of machine
learning researchers
and students to
bridge this gap to
bring active

inference into
its renaissance
to this end
I've been working
for the past
seven months on
sabbatical to
finish work on a
comprehensive
textbook
the aim of the
textbook is to
provide the tools
to bring active
inference to a
wider audience
primarily those in
machine learning
research and
applied fields such
as robotics
and to decrease
the challenge in
learning the
material largely by
separating it
for much of the
neuroscience background
that is usually
a prerequisite
this decrease in
prerequisites means
labs will have to
spend less time
helping students
becoming acquainted
with the field
and researchers

outside of
neuroscience will
find this book
an accessible
entry point that
uses terminology
familiar to
machine learning
rather than
neuroscience fmri
image analysis
all derivations
are in one place
currently in the
field many
derivations are
spread across
different papers
even behavioral
papers instead of
just technical
ones and it's
hard to know
where to look if
you want to
understand a
particular equation
or concept
part of the
success of deep
learning in the
last decade has
come directly from
focusing on narrow
improvements to
specific aspects of
the modeling
there are many

open questions in
areas of research
such as how to
prune policy
trees exploring
second-order
optimization rules
for state and
parameter updates
and scaling
active inference
the increased
accessibility for
researchers would
also lead to many
new industry
applications such as
autonomous vehicles
robotics video game
design and ai
the textbook would
also put a spotlight
on bayesian mechanics
and invite
contributors and
contributions from
researchers as
exciting nascent field
grows and develops
parts of the
part four is
largely a
literature review and
can be very helpful
to those writing
about active
inference from
fields such as

philosophy psychology
sociology and
many others and
the historical
context sections
that are part of
this book provide a
lot of that
context as active
inference is built
upon decades of
research in
neuroscience psychology
and many other
fields and also
draws upon current
work in many fields
that have emerged in
the last 25 to 30
years finally
LaTeX reproducibility
may offer interesting
ways to rearrange the
book and integrate it
with the code for an
online only experience
now i'd like to share
with you some of the
progress of my textbook
and the general
structure the textbook
is divided into four
parts the first part
introduces fundamental
concepts to set the
stage in particular i have
focused here on
presenting well-known

statistical ideas from
the perspective of an
agent modeling its
environment who states
it must infer from an
observed noisy signal
the second part focuses
on continuous and
discrete state space
formulations of active
inference where the
algorithm of focus for
the continuous state
space formulation is
active generalized
filtering part three
which i'll begin writing
in a couple of months
focuses on a sketch of
Bayesian mechanics and the
required background designed
with the knowledge this
field is still dynamically
changing and evolving
here i will focus on some
of the fundamental
concepts and ideas as
well as code simulations to
allow readers to get a
deeper and more intuitive
understanding of some of
these challenging ideas
finally part four is a
systematic literature
review that covers all the
various applications and
extensions to active
inference that have been

innovated in the last six to
eight years these
applications include things
like robotics all the
behavioral modeling and
neuropsychiatry and human
and animal behavior theory of
mind and so on the
extensions talk about how
applications of active
inference can be used to
talk about dynamic systems
more generally and apply to
things like ecosystems and
to economies and other
types of things like
governance and so on
as of this week the rough
drafts of part one and two
are complete and ten chapters
have been presented to the
active inference institute
in support of this textbook
are four separate tools that
i will discuss over the next
few slides
but before i do that i would
like to first highlight some
special aspects and features
of my approach to this
textbook
the major focus is on
writing this book for a
machine learning audience or
students learning in this and
adjacent areas
neuroscience is out of scope for
this book

many of the recommended and
even optional prerequisites
that are shown here are
typically known by
undergraduate students in
science and engineering
and certainly by graduate
students in these fields
nonetheless the book is
written to be readable by those
that desire to focus on
everything that is the math
the code the concepts
those that just want to focus
on the math but are not
interested in implementation
and even those that may skim
over the math and just try to
understand the ideas
intuitively
one thing that's very
important to me and trying to
express these ideas clearly is
by spending a lot of time
working on typesetting and
style
which is very important to
successful learning so i've
spent a lot of time attempting
to make my work clear and
readable
to this end i have margins
which collect specific key
terms for references later
which you can see in some of
these figures that are shown here
and these terms will eventually
correspond to the ontology

project ongoing at the active
inference institute
margins also provide further
explanation to accompany the text
and this will be useful to readers
who want more detail and
explanation
there's a large focus on building an
intuitive understanding of the
concepts
for example importantly all
algorithms are explained from
scratch
and this book we typically start with a
description of the agent environment
modeling problem
we then start the book with a univariate
case
extend to the multivariate case
then we introduce variational inference
we add dynamics
generalized coordinates where applicable
hierarchical models
action
and also learning and other modifications
we might add to our models
we have a very big focus on figures
clearly walking the reader through the text
and giving detailed visualizations of
important concepts
and in terms of how the textbook is set up
the early part of the book focuses on
basic concepts such as hidden state
estimation
that is estimating the conditional
distribution of a latent variable
given some observed data
the aim here is to explain the modeling

paradigm in the context of an agent
attempting to infer the states of the
environment
that is the interaction between a
generative model and a generative process
a perspective that differs from the
Bayesian inference style that is
normally taught
in universities and in many
introductory textbooks
typically introductory textbooks on Bayesian
inference focus on parameter
estimation or learning
and part one introduces the expectation
maximization algorithm as a way to
explain the connection and separation
between hidden state inference on the one
hand and parameter learning on the
other
additionally a large focus has been placed on
variational inference which is explained in
detail
and the book features a catalog of all the
different forms of variational free energy and
expected free energy in the literature and
how they can all be derived from one another

the book also covers raw and bollard style predictive coding terms and ideas such as key ideas such as prediction error minimization as well as clear and intuitive explanations of fundamental concepts such as surprise all the textbook focuses heavily on building intuition through derivation and the general flow of most of the chapters is to set up the problem that needs to be solved

which is to be solved which is to be solved which is to be solved which is defining an interaction between the agent and the environment showing the elements needed to solve it which are usually random variables and parameters that form a joint distribution or generative model replacing probability distributions with their algebraic formulations and

which is defining an interaction between the agent and the environment, showing the elements needed to solve it,

which is usually random variables and parameters that form a joint distribution or generative model,

replacing probability distributions with their algebraic formulations,

and then moving through the algebra to a final analytic or gradient-based equation.

The reader should be able to recognize most equations in the literature upon reading this book.

I'm also making extensive usage of Bayesian networks,

and other types and styles of graphical models, such as factor graphs, will appear in Part 4.

There are hundreds of custom figures that have been created so far,

and the figures are detailed to give the readers a deep understanding of the different types of content

that is covered throughout the books and also summarizes much of the information and equations

that are pervasive throughout the active inference literature.

Another important focus of the textbook is that many of the models that are presented are also shown in pseudocode,

which should aid the reader in implementation.

And finally, each chapter is filled with numerous what I call experiments,

and these experiments correspond to the Jupyter Notebook and try to show the application of a concept in a simulated environment.

So a lot of these experiments start out by generating data.

So we have some kind of generative process,

and then we have that data passed to a generative model or the agent,

which then attempts to either perceive and learn from that data and even act on it.

And the example that's shown on the right margin here,

this is just a perception problem on a continuous grid,

which has been divided into pieces for the purpose of the simulation,

but it represents a continuous state space.

And the agent in the bottom left corner, shown as a mouse,

has a prior belief about where some reward food is and its environment,

but it then needs to perceive from sensory data that it observes the true location of that reward or food,

which is obscured or occluded in some way by the mist that's shown in that figure.

So these types of experiments give the reader a better sense of how to apply these statistical ideas to a real-world situation,

so we can understand how it might apply to some kind of autonomous agent.

Next, I'd like to cover and shift my attention toward Jupyter Notebooks and videos.

And I'd like to note that upon publication of the textbook,

these Jupyter Notebooks will be released on GitHub and should be fully reproducible using Docker and other version handling tools.

One of the big emphasis on the Jupyter Notebooks is it has to be direct correspondence between the equations and explanations in the code and in the text.

This will build a direct understanding and show applications of the concepts explained.

Notebooks are filled with simulations and visualizations, many that appear in the main text.

And in addition to that, I have also over the last seven months been presenting chapter presentations to the Active Inference Institute.

So far, a draft version of 10 of the first 10 chapters of the book have been recorded at the Active Inference

Institute.

And these are just some sample slides that I've prepared that try to explain these concepts in great detail.

In the final stages of writing this textbook, video lectures will be re-recorded and released alongside the book.

I also plan to create detailed code walkthrough videos that walk through the different examples in the Jupyter Notebooks.

Now, I'd like to talk about a few planned future resources I'd like to work on after the book is complete or toward the final stages of the book to have further support and educational tools.

Some of these planned future resources include a software suite in Python to enable an alternative learning approach for those who do not wish to learn about the algorithms from scratch.

And this will expand the possible landscape of engagement.

As PyMDP already exists, I will not be working on a discrete state space Python package, but I'd like to fill the space for things like active generalized filtering and also just an availability of different types of simulations of Bayesian mechanics as it's currently defined today or at least the different versions and varieties of some of those key concepts.

I'm also very much interested in interactive learning and my aim would be to have these preset simulations concepts that are explained in text with various simulations interspersed in the form of plots and demos and other visualizations.

And the idea would be that the user could manipulate sliders and knobs to tweak various parameters that would help aid in learning as they get a feel for how these systems behave, especially ones.

Most of the systems we talk about are dynamics. So seeing how they change over time.

The Active Inference Institute has made tremendous progress in the past couple of years to provide a collaborative environment of research for researchers and for students of active inference.

I hope to be part of this ecosystem as I continue to support the spirit of accessibility and collaboration, and I'm excited to continue to contribute to this ecosystem of shared intelligence and look forward to what we can build together.

I would like to thank the Active Inference Institute for hosting my presentations, code reviews and feedback sessions, and inviting me to present the symposium and also thank my employer Kungfu AI for letting me take time off to write for the past seven months.

Please feel free to contact me at any time. Email is the easiest way, but I'm also available on the Active Inference Institute Discord.

If you would like access to the textbook and related materials, please send an email requesting access and I can get you set up.

And that's all I have for today. Thank you very much.

Awesome. All right. Thank you, Sanjeev.

Thank you, Sanjeev.

Greetings, Ines.

Hello.

Welcome.

Hi, Daniel. Thank you.

Yes, well,

we're really looking forward to your presentation. You can share your slides and go for it.

Great.

All right.

Amazing.

Can you see it?

Yep. Looks great. Thank you.

All right. Good.

So thank you so much for having me. I'm really happy to present this paper today, and particularly because it's the very first time that I present this work.

So I'm really looking forward to any feedback and any reactions to this work.

Okay, so the aim of this paper was to apply the free energy principle to analyze the interdependent dynamics between living systems on the one hand and non living planetary processes.

And this hopefully would, we hope, allow us to gain some insights into the future states of Gaia and by Gaia we're going to mean the Earth and formulate some predictions and devise potential interventions to alleviate future dysfunction on the planet.

Okay, so then the lineup for the presentation is I'm first going to go over in brief the theory, the concept of Gaia, so known as the Gaia hypothesis.

Then I'm going to go into just giving some essentials on the free energy principle, which I'm assuming that it's not really necessary, but just for the sake of the flow of what I'm doing, my argument, then I'm just going to present some things on the free energy principle and specifically on self organizing interacting systems.

And then I'm going to apply the free energy principle to the Gaia hypothesis in a sort of comparison analysis as in like how do they, if at all, complement each other for epistemic gain.

And then I'm going to, after having done that, this should be to the lawers to inform or to gain some insights as per the human behavior in relation to Gaia or the planet Earth.

Okay, so let's see how it goes.

So first I want to introduce the notion of biophilia, which plays a fundamental role in this work.

So biophilia came first or came more prominently in this particular book here.

And it's quite interesting.

The idea is that the relation of humanity to the rest of the natural world is essentially one of dependence.

And this is very much already in the literature on the free energy principle.

But more than that, there is this view of the necessity of a continued survival and flourishing that is utterly contingent upon the state and functioning of the environment.

So we're talking about things like very utilitarian things like survival, flourishing, mental health, right?

So individuals interact with the environment.

So to achieve that.

But then on top of that, we wanted to add another conceptual layer, which is of concerns how humans possess an innate cognitive disposition towards nature, which is known and developed in this particular other field of work as biophilia.

And this is the idea of natural psychological disposition of humans to seek out and connect with other living organisms and the natural environments.

So that's kind of like the idea that is encapsulated into the whole framework and theory under biophilia.

So then this seems to, if that's the case of as things are described by biophilia, then that seems to be the case of biophilia.

And that seems to bring us to a dilemma.

On the one hand, we have this natural tendency, so to speak, of humans to engage with the natural world in a very positive and constructive way.

And on the other hand, we see the lack of action to protect the earth.

And here we have in mind, particularly the case of the climate crisis.

So in the presentation, I'm going to make my way slowly towards getting to that and to better understand the dilemma.

Okay, so let's go in brief over the Gaia hypothesis.

So this is a concept that comes more prominently in James Lovelock's book, Gaia, a new look at life on earth.

And the idea there is that 3.8 billion years that life has persisted on earth and conditions have remained remarkably constant and favorable across a huge range of parameters.

The Gaia hypothesis is that this consistency or constancy to be is a natural product of evolution.

In a modification, however, of the canonical Darwinism, the Gaia hypothesis somewhat modifies canonical Darwinism.

How?

Well, by thinking that rather than evolution of the individual organisms on an inert environment, what evolves is the whole earth system with its living and non-living parts existing as a tight coupled entity.

So this is the kind of like, so to speak, refreshing or new way of thinking that somewhat reformulates a little bit of Darwinism, right?

It's not the emphasis is not how one's individual organism evolves or species evolves, but the whole earth as a system.

So that's quite interesting cause for example, the, the view that the combined genetic wealth of information of the whole system becomes much more prominent here than the Darwinistic kind of view.

And these acquire characteristics that are conductive to its continued existence and in turn affect changes on the environment.

So then seen as a whole organism or as a whole system, one can look at Gaia or the earth as a complex system, which appears to have the goal of regulating the climate and the chemistry at a comfortable state of life.

And these views and these ideas have come to gain more official expression in 2001 with the Amsterdam declaration on earth system science precisely endorsing this view on the planet earth as a whole system.

So then the Gaia hypothesis is a complex system in, in, in, in various ways because it changes adaptive in an adaptive and non-linear way.

It also displays dynamical image emergent and sudden tipping behavior.

And when bifurcation points are reached, the new tractor states form, um, new tractor states are formed.

And this means that it also self-organizes and there is interdependence between the processes of the system.

And finally, the Gaia hypothesis seems to be quite interesting and extraordinary in the sense that it brings together so many different fields such as ecology, climate science, cybernetics, complex systems theory, chaos mathematics, and philosophy of mind.

And that's just a few.

All right.

All right.

So then, um, the idea is of the Gaia hypothesis is that we could look at the planet from orbit, or as it is known, the overview effect.

And the Gaia hypothesis offers this glimpse of a fundamentally different way of thinking of the planet of thinking that in fact dominated mostly, mostly by human history.

So it's a little bit of like, um, that, um, very inspiring clip by Carl Sagan, the pale blue dot whereby the world, uh, is considered.

Um, so throughout the human history, there's typically a tendency for the world or the planet to be considered from a single living organism, uh, though, rather than an it.

So that's, that's the idea of overview effect and they taking this overview effect to look at the planet as this whole living system rather than an it.

Okay.

Okay.

So then we can think about the Gaia hypothesis as like Newtonian physics, or even as the free energy principle.

The Gaia hypothesis can be seen much more as a framework or a method of investigation rather than a falsifiable empirical claim.

So in this sense, these particular frameworks, they are, um, and should be judged by their usefulness and not by their purported, um, um, uh, correctness.

Where usefulness, importantly, usefulness is to be determined by factors such as parsimony, coherence, and consistency, plus it's, um, empirical and epistemic power to explain phenomena, produce new facts, and so on in a given context.

And at a given time frame.

So that's the stance on, uh, that we think that should be taken on the Gaia hypothesis.

All right.

So now having said a few hours on the Gaia hypothesis, now, um, I'm going to just prepare the stage by saying a few words on the free energy principle, which, uh, given the topic of, uh, this conference needs an introduction, but for these purposes, I must mention just a few things.

So the free energy principle, we take it to be the case as well, that it is methodology to understand, um, over served patterns of behavior, uh, sorry, observed patterns of behavior in the natural world.

So then we apply the free energy principle as the tool or methodology to analyze these interdependent dynamics between living systems and planetary processes, right?

Kind of like this goggles, this methodological, technical, scientific goggles that you put on to gain insights into the future states of Gaia or the earth, uh, such as to make future formulate predictions and devise potential interventions in this particular case with the care of, um, alleviating future potential or present, um, actually dysfunction.

Okay.

Okay.

Okay.

So the free energy principle, um, very much align aligns with, um, with this, uh, thinking by Ewan, uh, Schrodinger, uh, and the view that the second law of thermodynamics or under the second law of thermodynamics that open systems like us should tend to dissipation.

However, interestingly, there are displays of this super cool thing, the negative entropy, which are pockets of the universe where order and bodily integrity are maintained in the face of the search towards chaos or towards entropy or towards dissipation.

And this view of pockets of the universe is incredibly interesting, um, because this is where we find self-organizing systems like us.

And what is so peculiar that distinguishes this self-organizing living systems from non living systems is that they have the capacity of acting upon and exchanging matter, energy, information with the environment in order to seemingly defy the second law.

And therefore, or thereby avoid dissipation, uh, and, um, chaos and entropy.

So that's extraordinary.

So then active inference, um, becomes this very useful tool to, uh, tell us how to explain elucidate.

How, um, these systems seemingly defy the second law, how this, uh, they resist entropy, how the self-organizing systems combat entropy through the interaction, maintaining their integrity via homeostasis.

So then they maintain this energetic integrity.

They utilize energy to stay within a specific range of states that is vital for their survival and preserving their existence.

And they also engage with these feedback cycles, which are the feedback actions, uh, like drinking when thirsty or moving away from heat, um, signify working accomplished by bound energy contrasting with the free energy.

They also define the thermodynamics, as I just mentioned by seemingly challenging the law.

They minimize entropy and safeguard their free energy.

And finally, they, this behavior that we observe in this self-organizing living systems, um, in striving to minimize free energy, um, is, uh, or agrees with a certain principle that we know, which is the free energy principle.

Okay.

So then, um, as I said, um, earlier, and as we know, but just, just to keep in mind, then active inference becomes a very useful tool to understand and make predictions about these coupling dynamics taking place between interacting systems.

The living systems internal states in both is both operationally closed and thermodynamically open to the external states, creating this distinct boundary, which we usually mathematically capture, uh, with Markov blankets.

And then disinteraction unfolds through the military, uh, or active and sensory states.

Then the key point here to keep in mind is that what we want to have for adaptation and wellbeing is that there is a balanced interaction between sensory and active states.

And this is because it is in within the interaction between sensory and active states that, so to speak, all of the important things happen in order for the system to be well adjusted, attuned, adapted, uh, to the environment.

So this is extremely crucial for the point that I want to make, uh, later on.

So now, um, I want to apply the phenology principle to the guy hypothesis and see how that, um, comes, uh, together.

In identifying and developing.

And why do I want to do that?

I want to do that because I've also already mentioned it, but I'm just going to, uh, refresh it because I want to identify and develop interventions to rectify any potential maladaptive responses.

Mainly actions which are antitentical to maintaining the integrity of the system.

What we want is to have healthy, well-adjusted, adaptive systems.

And human behavior, particularly that leads to the imbalance within the Markov blanket states is, or could be considered sometimes maladaptive or pathological.

And this will be the last point I'm going to make in the presentation, but for now, um, want to focus on how does the Fianchi principle, uh, deal with the guy hypothesis.

Uh, as a self-organizing system, like we were just saying, the guy who meets the requirements to be formalized as part of an active inference system.

It exists in a non equilibrium, equilibrium steady state, which means that its processes and states must persist over time within a range of states that are far from equilibrium.

That means resisting entropy.

And the Gaia is also distinguishable from its larger scale cosmic environment, such as other planets and the rest of the universe.

It is also defined by the conditional independence that we were just talking about between the internal and external stage, which influence each other only vicariously.

So then we can, uh, taking that into consideration, we can even see that some of the people have already done, uh, these kind of like line of reasoning.

They're already running through, and they did that by employing Markov blankets to analyze the earth's climate system dynamics, which is quite interesting.

What they did was they looked at metabolic rates of the biosphere, uh, and they represented them as internal states and space weather as external states.

And then they looked at active states as mirrored greenhouse effect shifts and sensory states as echoed ocean driven temperature changes.

And then internal and external states, uh, the idea here is that they indirectly influenced each other through oceans delayed response.

Then earth's climate is interpreted as anti anticipatory minimizing variational free energy.

So we take these work and we want to take it a step further.

Um, agreeing with this work and building upon this work on which shows the climate change.

Um, we wanted to look at, um, the more broad understanding of Gaia through the lenses of the free energy principle.

So, uh, instead of focusing on climate change, which is very brilliantly done by these researchers, we wanted to then scale it up and apply the free energy principle to understand, um, the Gaia.

So then one can think of the internal states as the biosphere geographical processes and the external states as everything that, that is beyond the system, such as the outer atmosphere, siliceous bodies, and molten core.

Then we can think of sensory states as the external environment, uh, impact, such as the earth rotation, orbit, tides affected by the sun, moon, planets, plate tectonics, climates, weather influenced by solar radiation, space, weather, et cetera.

And then we can think of the active states as earth adaptability and adjusting properties from external influence, such as gravitational force, alters, orbits, magnetic field interacts with solar wind, or emission of detectable radiation influences distant objects.

So that's a few examples.

So then I want to focus on the active states.

Active states then through these lens active states then embody its primary mechanism for self-regulation.

Representing its efforts to remain within the bounds required for ongoing existence based on predictions about the present and future states of the system.

For instance, the current levels of greenhouse and albedo effects, soil pH, and the number of living organisms reflect the predictive attempts of the Gaia system to adapt to anticipated effects of solar events, volcanic eruptions, ocean salinity changes, and other environmental factors in the future.

So it is in this context that the role of life itself assumes a paramount importance in the self-regulation of the biosphere.

Because the role of life itself, and particularly as we are going to focus on next, the human life has a particular influence on all of these sort of already natural dynamics going on in the Gaia.

So this allows us, before we move into the human effects, this allows us to make one final point before we move there, which is to address the criticism that has been made in this paper by Vicente Haagha on the Markov blanket trick.

And what they have identified is that a purported witness of the free energy principle.

What we say is that a witness of the free energy principle that is identified in this paper actually speaks in favor of its application to the Gaia hypothesis.

And let me just try to explain that.

So, Gaia and colleagues, they rightly point out that Markov linkage do not automatically capture every relevant property of biological systems.

Relational properties, relational properties such as affordances, relational properties such as affordances, and by affordances, they mean what the environment offers the animal, what it provides or furnishes either good or evil, fall into this bracket.

So what they're saying is that Markov blankets do not automatically capture every relevant aspect of biological system such as affordances.

So we want to point out that however, affordances correspond to Gaia as one rung in the lather of the nested Markov blankets within the system in question at every level ranging up from say, cells, organs, animals, ecosystems, solar system, the whole cosmos, and how is it that we think that we are think that this particular framework that we bring here does have or solve or address the notion of affordances. Our solar system is one that affords a living planet. So the affordance available is an environment in an environment are not determined solely by the physical properties of the environment. And we think that this is the point that we would like to address in Haha's paper is that the affordances are not determined solely by the physical properties of the environment, but also by the organism's goals, abilities, and previous experiences. And in this case of the sun. So what is meant here is that the sun is what has the affordances for or what affords anything else to happen on the planet at whatever each scale, right? The physical properties of the sun, as well as the goals and abilities of the organisms on Earth, shape the affordances that the sun provides. So it's a quite interesting way of thinking that any affordances whatsoever that we might think of that we have as living systems, they are afforded by the sun. So that's the idea.

Okay, so now having gone through how the free energy principle somewhat speaks to and contributes to understanding the Gaia hypotheses. Then we want to focus on how human behavior has a profound effect on the dynamics that already existing in the natural world. Okay. So according to the free energy principle, living beings have a natural tendency to act upon their environment to prevent dissipation of their integrity. So that's what we do in order to remain alive. There are therefore two ways for a thing like us to minimize its free energy and thereby maintain our own integrity. Upon evidence, as we navigate the world and gather and observe evidence, we have two options. One is to change the model. The other one is to change the world by acting upon it, like I just said before. So then why is this relevant?

Well, this is relevant because there is the possibility that in the ways in which living systems and in this particular case, human beings interact and adapt to the world, there is the possibility that we see maladaptive behavior. And how would that be? Well, if it is the case that there is a rejection of the information available so as to remove the imperative for action, or if it is the case that we are possessing the information available, but we still remain or refrain from taking action, then there seems to be if the free energy principle is correct, if the free energy principle is to be correct, that systems, living beings tend to act upon the world in order to remain alive. And if it is the case that one rejects that kind of information and does not take action to remain alive, such as the condition of the climate crisis, then we might be looking at maladaptive behavior. So if the free energy principle holds, then the observed contradictory behavior is a maladaptive state or should be seen as a maladaptive state that arises from an imbalance within the Markov blanket. And how do we see this imbalance within the Markov blanket? We see it to put it in very, very simple terms, we see it as an insulation of internal states that are not being affected by the information available outside of the system. So the influences or the interactions are occurring between internal, active and sensory states. And this is an imbalance that results in an insulation or an echo chamber effect in internal states that are impermeable to the information from or potentially existing in the external states that demands action.

So then some food for thought. What can we make of this? The free energy principle, if we are to use the

free energy principle to model the dynamics of the relationships between humans and the environment, then we think that it is possible to see the current lack of pro-environmental behavior as a pathology of the human species that must receive treatment. We label this condition as biophilia deficiency syndrome. Biophilia deficiency syndrome, and we use, it is important for us to focus on that to learn some constructed, to construct some validity, is useful to consider in this case other applications of the free energy principle that may be useful as an analogous case of pathology mainly conducted in psychiatry and neurology. And here, and very, very, very short because we don't have time, the idea is to see it as a failure to assign the right way or prediction to prediction errors. So then the free energy principle, from this point of view, this is a proscenious pathology because doing nothing is a base optimal response in the face of pathological attention to various sources of evidence. So evidence is ignored. So concluding and recapping, according to the free energy principle, living systems possess a biological encoded inclination to interact with the environment in order to survive and adapt to the environment. And despite the climate change being a pressing issue and the information being available, the evidence being available, proactive measures to reverse its effects are not giving the priority they deserve. This state of affairs seems to show very clearly a divergence from a natural inclination that we call biophilia or that it's existing in the literature as biophilia. This discrepancy between the expected and the observed actions can be understood in the framework of the free energy principle or so we think as a newness. It implies that systems, including human societies, may form erroneous inferences that prevent them from effectively addressing the problem at hand. Now, thinking about future directions, and this is my last slide, current ecological imbalances could be thought of as a stuck state that require an active disturbance to bring about some change. If not, a stuck state is a state that a system is at a local minimum where the system is at and won't get out of that state unless it is disturbed. So that's the principle in complex systems theory. And we think that that's how this particular state should be looked at, but that's for future research. And what we would be calling is we would be calling for amounts to the kind of homeostatic awakening. So the sort of like getting out of this stuck state would require some kind of like homeostatic awakening, a deliberately induced disruptive shift in the trajectory reflecting a priorization of the planetary homeostasis over infinite growth or staying in the same place. And yeah, this is my last slide. Thank you so much. I would like also to acknowledge my co-author, Casper Montgomery. Thank you so much for your attention.
Yes, thank you. Well, a lot of ways to go. I'll just ask one question. Just in the years that you've been in the game, how do you see the active inference ecosystem and the ways also that it's being applied to ecosystems evolving? Like over what timescales or in what ways does that really present itself to you?
Slow. Sadly, slow. I think there's a lot of work that needs to be done. We try to do this particular one here to apply, employ active inference to the planet or the earth as a living system precisely to bring out or to overcome. So one is to look at this problem of the climate crisis as what we think it is, which is more of a pathology. If the free energy principle holds, then it seems that there is a lot of work that can be done and that would very much be useful in the climate crisis in general.
There is some other work that came out recently by Maxwell and colleagues as well on the ecosystems,

which is really, really cool. And I would also mention that one. And I really hope that we can pursue this line of research and employee active inference in the free energy principle to this particular case. One of the things that really attracts me to the free energy principle is the capacity that it has to bring to the forefront certain laws of physics, but also the living systems navigating and interacting with their environments, which I find much more appealing than any computational framework per se. So that's what I find that it's a very useful conceptual toolkit that I think that I would really like to see more applied in the future for the climate crisis on the earth.

Great. Well, thank you so much for building on that line of work and for sharing with us tonight.

So thank you so much for having me till next time. See you. Thank you. Bye.

Bye. Thanks.

All right. Well, great talk by Ines. And now, welcome Aswin. How are you doing?

Hi, Daniel. I'm good. Hanging out. I'm looking forward.

Yeah. Yeah. Yeah. Sounds good. And yes, looking forward to your workshop on sophisticated inference in PyMDP. So it can be up to 90 minutes or it's totally okay if it's less and we can take a short break.

But please take it away and just let me know however I can help.

Great. Thank you so much. Maybe I'll share my screen and start. So regarding the structure of what we're doing here,

I'm assuming that people who see this later might want to try it hands on along with the tutorial or something.

So that's the structure I kept in mind. And so I'll go slow. Please bear with me.

So welcome all of you to this session on sophisticated inference in PyMDP. So here we are going to attempt to model some sophisticated inference simulations, especially the one in the original paper using the PyMDP module and how it is not part of PyMDP right now, but we are on the process of adding sophisticated inference to the PyMDP module. And I'm going to mainly talk about the code that I have developed to kind of add that functionality. Right. So I'm Ashwin Paul. I am finally a PhD candidate at Monash University. And mostly I work with active inference models and try to understand how to use them as an explainable model to basically understand emergence of intelligent behavior.

Right. So let's dive into the material right away. So to give an intro of free energy principle, I'm sure all of you right now have an understanding of what it is. But the central idea is that an agent is always trying to minimize the entropy of its observations. Right. So if an observation is having really low probability in your mind and that happens, then you are probably surprised and vice versa. Right. So and the entropy here is defined as the information theoretic entropy, where if you have a low probability, then automatically this is a high surprise or a high entropic observation. And as we all know, active inference also gives us a methodology to define what we call an agent environment loop.

And this lets us define what is the agent that we are looking at and what is the behavior of the agent given the environment around it and so on. Right. So you're also familiar with the idea of Markov blanket.

And this is important because we always have to remember, I mean, have to remember the difference between the generative process and the generative model, which is quite a famous point of confusion in active inference literature and for the people who try to understand it in the beginning. Right.

So the central question is that how does an agent minimize entropy? Because how does an agent know which observation is low or high probabilistic? Right. So that is by maintaining a generative model. And the generative model will tell you which is a high probabilistic observation and which is a low probabilistic observation.

So the idea is that all the agent has access to is an observation that is coming from a generative process, which the agent cannot directly observe. And an intelligent agent will try to build up a generative model in its mind,

which is a model of the hidden states and the observation it has access to. And it can hope to kind of compute probabilities using this generative model. Right. But there is a problem that in general, it is an intractable problem to kind of marginalize the probability of observation from the generative model.

And that is why we have to define an upper bound on the surprise that the agent is trying to minimize. And there comes the idea of free energy. Right. So this upper bound, the agent is supposedly minimizing is the free energy. And that's why it is the free energy principle.

And here we have a new term called capital QS, which can be interpreted as the belief that the agent is maintaining about the hidden states in its generative model.

And this quantity is the free energy. And traditionally we see this variation free energy being interpreted as in mostly the machine learning way, where it is a balance between complexity and accuracy of the model.

So when minimizing the free energy, the agent is trying to come up with a simple model, but at the same time, an accurate model, because here it's a minus sign with accuracy. Right.

So it can also be interpreted in the physics way, where the agent is always trying to minimize the energy of the model, but at the same time, maximizing the entropy of the model, which goes in conjunction with the maximum entropy principle and so on from the classic literature.

So given that this idea of the model is the first thing that you might want to define a generative model for the agent, which is informed or not informed depending upon the experiment that you're trying to model.

So in classical active inference, usually decision making is defined in terms of policies.

So for example, if you are an agent in this environment, so in the Mario game, Mario is the agent and everything else is the environment.

So in the environment and Mario has three available actions run, jump or stay in this environment.

And the classic definition of policy is that it is a sequence of actions in time.

So if you have a time horizon of capital T, then a policy is nothing but a series of actions you might take.

So run, run, run, jump and so on in time.

So this is the superscript is the action.

So here it is jump, run and so on.

And the subscript is the time.

And then what you can have is the policy space, which is a collection of many such policies, smaller pie.

And what you essentially do to take decisions in active inferences, compute, not optimize, compute the expected free energy of every policy in your policy space.

And basically that can be interpreted as a balance between risk and ambiguity.

And yeah, so when you compute this expected free energy, what you're trying to do is minimizing the risk.

That is how different is your belief about the observation from your prior preferences, capital C.

So this is also part of the generative model when you're trying to model control.

And at the same time, you're trying to minimize the ambiguity when you're choosing a policy that has the minimum expected free energy.

Right. But this formulation has a problem that a policy space quickly becomes intractable, that there can be an enormous number of small pies or policies in your policy space and sitting and computing the expected free energy for all such policies.

Even for a small time horizon is not possible.

But this is the classical structure that has been implemented in PIMDP.

Nonetheless, where we have different modules in PIMDP that is meant to be implementing different aspects of behavior.

So, for example, for inference or perception, we have belief propagation, fixed point iteration, marginal message passing and all that.

So, we have different methods implemented in the inference module.

In the control module, we have different methods to evaluate expected free energy for policies.

One, depending upon the expected utility.

The other one, depending upon the classic method that I just explained.

Then we have a module for learning.

So we learn the parameters in the form DP like capital A, capital B.

So likelihood, transition dynamics and so on.

And then we have algorithms for implementing all this in the algorithms module.

And then the most powerful thing in PIMDP right now is the agent class where it is easy for you to kind of define the agent environment loop.

And we are trying to build up.

So today I'm going to talk about an agent class that implements sophisticated inference rather than the classical active inference that we just saw.

So, as I mentioned, how many valid policies can be defined say for a time horizon of 15 in classical active inference?

Right.

So the first policy, of course, is a series of first action that is jump.

Then you can have the last last action changed and you already can see that there can be n number of combinations.

And for this simple case, the policy spaces as biggest 10 to the part 13.

And in a stochastic problem setting, there is no way to kind of come up with a small subset of this policy space so that you can tackle this problem of computational complexity.

And yeah, as I mentioned in a stochastic problem setting, it is an intractable size policy space.

There comes the idea of sophisticated inference where we are thinking about taking this instance in a different way.

Right.

So rather than thinking about sequence of actions in time, we can directly think of what to do when we see something depending upon our beliefs about the current state and beliefs about the future.

Right. So if I see myself in a current situation, what should I do?

And that's more like a straightforward thinking of how to take actions.

And here the expected free energy, the structure of the expected free energy is the same, but we are not evaluating expected free energy of policies, but expected free energy of observation action combinations.

So if I see something and if I do this, what's the expected free energy?

And that's what I'm trying to minimize.

That's what I'm trying to optimize in this setting.

Right. So here again, we have the risk term where we are trying to minimize the deviation between the belief and the prior preferences.

We also have the ambiguity term and this together makes up the expected free energy of this time point at time t.

But we also have an expectation about what's the expected free energy in the next time step.

And to evaluate the expected free energy of next time step, you will have to again compute this equation with O t plus 2.

And for that, you will have to again compute this equation with O t plus 3 and so on.

And this automatically becomes a tree search because of the recursive way this equation is defined.

And it comes with its own problems, but there are clever ways to get around them.

And that's, we are going to discuss that in the code today.

So given the structure of sophisticated inference here, as I mentioned, the tree search replace the policy space that we saw for the traditional active inference we are used to.

So in this workshop, what I am focusing on is how to kind of define a generative model and given an environment.

So for example, this is the grid that is stimulated, simulated in the original paper.

And we are going to talk about how to build up a generative model for this grid that can be used in the sophisticated inference in the PMDP module.

And so basically what I am trying to talk about is that the environment will have a step function that takes an action from the agent in PMDP.

And the agent will get an observation out of that action.

And we are, we'll talk about this particular function, agent dot step and agent dot steps will step will take up an observation and try to give, come up with an action for the next time step.

Right. And this creates a loop and cleverly designing this loop will let you see emergence of purposeful behavior in the sophisticated inference setting.

So for example, in this particular grid with sufficient planning horizon, you will be able to see that the agent is capable of navigating in this grid and so on.

So this is the example that I'm going to focus on in this talk today.

So I want to, excuse me, I want to right away jump in to the code.

So we have, um, the PMDP home, which I am, I'm hoping you are familiar with.

So we have this, uh, GitHub repository where we have the PMDP module and inside PMDP module, um, we have several, uh, parts for it.

So here we have the agent in the original PMDP module that implements, uh, the so-called classical active inference.

We have several, uh, environments, uh, and we have helper functions like learning inference maps and so on.

So this is the module of time DP, but in the, um, parent folder, we also have examples, uh, where there is, um, tutorials about how to use the agent class, how to kind of, uh, deal with the environments and so on.

So, uh, if you look at the full requests, so we are right now trying to, uh, merge sophisticated inference into the original PMDP module.

And today I'm going to talk about the code in this full request.

So if you are, if you want to try this, uh, hands on, you might want to go to this page, uh, where this full request is there.

Uh, and it has the same structure, uh, of PMDP.

It's basically, uh, designed using PMDP.

And here what we have additionally is an agent SI, which is a sophisticated inference agent, which does everything and also planning and decision-making in the sophisticated inference way.

And in the parent folder, there is also an example folder, uh, for sophisticated inference demo.

And what I am going to do today is to walk you through this tutorial of sophisticated inference.

And on the way I'm going to discuss, uh, and at points, uh, where I reference, uh, the helper codes, I'm going to go to that code and try to explain what's actually happening and how we complete that, uh, agent environment loop where we can see, uh, that purposeful behavior.

Right.

So yeah, so that's the PMDP home.

Then I also talked about, uh, the pull request.

So let's right away go into the Jupyter notebook.

So this is my local copy of this repository.

So it's easier for me to run it and show it in my personal computer.

So this is the parent folder with PMDP and examples.

And in inside examples, I have a demo folder for sophisticated inference.

And this is the notebook I'm talking about, right?

So here, uh, in this example, what we are trying to do is, uh, deal with this particular grid world task from the original sophisticated inference paper, uh, and make this agent, uh, or enable this agent to navigate to this, uh, red dot, which is supposedly the goal state in this particular task given a prior preference like this, right?

So this prior preference is quite informative in the sense that we right away can see that this

is the most preferred state, the white color and the surrounding states are kind of less preferred,

but more preferred than the far away ones.

Right.

So this is the grid world task that we are trying to use.

So the first cell is importing, uh, all the necessary libraries, uh, and some useful libraries like numpy and map plotlib, uh, and the most important one is PIMDP.

So I'm, I'm actually now calling the local copy of my PIMDP with the sophisticated inference implementation, not the original one, which is not merged yet.

So, um, and the first thing I want to talk about is the environment itself, right?

So the environment dot step part, where if I get some action, how does the environment work?

And for that inside this folder, I have an, I have a file, which is great environment, uh,

SI dot PY.

Uh, and this is basically an environment class.

So don't worry about how this environment is actually implemented.

The only thing to worry about is, um, this function that we are going to use, which is, um, environment dot step.

So this function will take an action into it.

And depending upon, uh, the current state of the environment, it will calculate what is the most probable next state, um, given this action from the agent.

So that's the idea.

And then it will also calculate a reward of, um, some, uh, a negligible negative value.

So this is the, uh, the current state of the environment.

If it is not the goal state and if it is the goal state, it will give a reward of 10.

And that's how the environment is designed.

Uh, and it will update the current state to the new state.

And basically what it will return is a new state, depending upon your action, um, the reward

for that action and whether it is an end of the episode and so on.

So this implementation is, um, the standard open AI environment implementation.

And this is the environment dot step function, right?

So in this grid, for example, uh, if I am right now in this state, if I take an action up, so

I have four available actions, uh, north, south, east and west.

So if I go on north, then the environment dot step will make sure that I am in the state that

is about the state.

And if I go east or west, then I'll stay here or south, I'll stay here.

So that's the idea.

And yeah, and there is an episode length limit that is eight here.

That means that I am restricting the, um, length of every episode to be eight, which is the ideal

length of preaching this gold state just to avoid confusion.

So in this environment, after eight actions, the environment will terminate.

Uh, and if you have to kind of reach this rule state in the optimal time point.

So that's the idea how the environment is implemented.

Okay.

Uh, I hope that is clear and there are many helpful, uh, functions in this environment,

like, um, rendering, uh, the environment, rendering a prior preference matrix in this environment.
So if you design a prior preference, this environment can, um, show that in, in, in a pictorial way, how your prior preferences, uh, that you will see in the notebook below.
So now it's time that we define a generative model for the sophisticated inference agent, right?
Um, before that, let's define the structure of the generative model that we want the agent to have in its mind, uh, which is tailor made for this particular environment, right?
So here in this particular grid world task, we have, uh, 25 valid states, um, starting from this state, all this black states in this path are valid states.
So there are 25 valid states.
And then there are four available actions for the agent, north, south, east and west.
So this is part of our generative model.
This is also in, uh, alignment with the reality of the grid.
Uh, but this is about what the agent has in its mind, right?
And then the observation is, um, just the state space.
The agent is, uh, it, the problem is fully observable.
So there is no ambiguity there.
Uh, then we define basically the number of states.
Uh, then we define basically the number of states, which is a list of, uh, your state space, number of factors, which is now one, because you only have one hidden state factor here.
Then number of controls, which, which is going to be four, uh, that's four available actions and your observations space like that.
So this is the structure of your generative model.
And let's look at, uh, the structure of the parameters now inside of POMDP.
So the first one is the likelihood function, which is, um, often denoted by capital A.
And here it is a function of how many observation modalities that I have and how many state modalities I have, right?
So if I run this cell, yeah, I have to run the parent cells to make sure everything works.
So I have rendered the environment, um, the structure of the generative model.
And here I have the capital A matrix, uh, which has a structure 25, 25.
That means that I have 25 states and 25 observations.
And here, because it's fully observable, uh, I am initializing it as an identity matrix of size 25.
So that's my likelihood, uh, matrix that I'm initializing for, uh, this particular grid world task.
Then the second element is the transition matrix.
So there, uh, so please note that I'm using all the existing PIMDP, uh, functionalities to define a random A matrix, uh, and then using an identity matrix, uh, on top of that.
So, yeah, I'm not doing anything new here.
It is the existing PIMDP functionality.

Then what I can do now is define, uh, the B matrix, which is also called transition matrix.

So the transition matrix and codes transitions, like where I'm going to end up in the future.

If I start from a particular state and take an action.

So that's the idea where it depends upon the number of states, which is the hidden state modality and number of controls.

So it has the structure of, uh, state action state, where if I take an action from a particular state where I'm going to end up.

So that is also a future state, right?

So I'm going to initialize it as the true environment state.

So now this is part of the environment that I've built.

It will give out the B matrix.

Um, it might be worth it to look at the structure of this B matrix.

So here, um, we have 25, 25, 4.

So that means that if I take an action, uh, from, uh, a particular state where I'm going to end up and we have the true, uh, transition dynamics for this particular grid by design.

So there is a function called get true B and that will give us, uh, the true B of the system, which can, which the agent can use.

Okay.

So ideally we would want the agent to learn this, but for the purpose of this demo, uh, we are assuming that the agent already knows the structure.

And then, uh, comes the prior preference, which is, uh, interesting here in the sense that it is defined as, uh, how closer you are to the goal state.

So if you are at the goal state, then, uh, clearly that is the most, uh, sort out state.

You prefer that the most, uh, and, uh, how, how do you prefer the neighboring states, right?

So that is dependent upon the square, uh, root of the distance or basically the distance from that, uh, particular goal state.

So you define a grid, um, which is eight cross eight, the same size as this particular grid world task.

Uh, and then we have, um, uh, method to kind of add values, uh, which is the preference you have for every state.

And if you render, uh, the particular C matrix, you can see the structure, which is the same where this goal state is more preferred and the surrounding states less preferred and so on.

So now we have, uh, the C matrix also defined in the classic PIMDP way.

And then, uh, I initialize that C matrix as the C matrix we evaluated in the previous cell, which is small C, uh, this particular C matrix.

Okay.

And then lastly, uh, for the generative model, we have capital D, which is your prior overhead and states.

And for that, I'm using a uniform object array.

So that means that I don't have a prior of where I am starting.

So let me run the bending cells.

So here the D matrix is a uniform distribution over hidden states.

I don't know where I'm going to start the simulation.

The simulation and so on.

So this is the basic structure of the generative model.

And then, um, we have the Asian class, which I want to discuss separately, uh, like the environment, right?

So given these, uh, environment parameters, how would you expect the Asian class to work?

So where is the Asian class inside this folder structure?

Uh, inside, uh, the PIMDP module folder, we have an agent SI dot P Y.

Um, which is basically a class again.

And similar to the environment class here also, we have a step function where this will take,

uh, an observation in, to the function, uh, and also a flag, whether or not to learn the environment, which, which is optional.

Uh, so if you disable it, it will, it won't learn, uh, the generative model.

If you enable it, it will update the parameters of the generative model.

And what basically it does is it will return an action, uh, which is to be taken at this point of time.

And the environment can basically use that action, right?

So in this file, we have the Asian class, which I will explain in detail.

So I am basically importing that Asian class in the cell.

And then we are going to try and reproduce this behavioral result from, um, the original substrate inference paper.

Okay.

Uh, and for that, so what we expect is that given this, uh, prior preference structure, there are local maximus in this prior preference structure.

So if you start from this particular point, if you do not plan deep enough, what you will end up is in one of these local maximus where you don't see that, uh, there is a highly preferred observation, say four steps down the line.

So if you are in this particular state, what you will see is this local maxima and you will go and sit there because the neighboring states are less preferred.

And this state, which is more preferred is not accessible because of the wall, uh, or the wall structure.

So you have to take a turn and pass through less preferred states and you need deep planning in order to enable the agent to do that.

The agent should be able to kind of simulate, um, four time steps ahead in time to see that there is this, uh, highly rewarding observation coming to kind of do that, uh, actions.

So that's the point that we are trying to, um, see in this particular demo.

So for, uh, low planning depth, uh, it will basically get stuck in one of the local maximus, but with sufficient planning depth, it will navigate, uh, to the goal state.

So that's what we are trying to see, right?

So we have different planning horizons.

And what we are basically doing is give the agent, uh, a generative model, which we right now defined the A matrix B matrix C matrix B matrix.

Then we have the planning horizon of capital N.

So here I am, um, iterating over planning depth.

So N will be one, three and four for the loop.

Then we have action precision, which is often denoted by alpha in active inference literature.

So that determines which action is to be taken.

So a highly precise action precision means that it will stick to the action with the lowest, um, expected free energy.

Uh, but a lower accent action precision is kind of probabilistic where it will, uh, also consider other actions.

Then we have a planning precision, which is part of the planning function.

We'll discuss, which is often denoted in the literature as gamma.

Then we also have a search threshold, which is extremely important for sophisticated inference, because as we saw sophisticated inference is a tree search.

And, uh, tree search is bad in the sense that it can have a lot of computations, but you have to define a threshold to kind of ignore many possibilities to make it work.

And that's the idea that we will also discuss.

Okay.

So just a preview before we go to the agent, um, class, what we are trying to do is in a loop,

we are going to, um, call the agent dot step and environment dot step in series.

So the agent will see an observation.

It will take an action.

Uh, that action will go into the environment.

The environment will give it back new observations and this loop continues.

And we want to see over time how this loop, um, evolves into a purposeful behavior.

And, and if we, if the agent at all is capable, uh, for that.

So before I reveal, um, the results, let's discuss the agent glass.

So in order to give an action, uh, when an observation is given, uh, the agent should have the planning and so on.

Right.

So here, um, is the agent, um, the sophisticated inference agent, uh, where we are actually using the existing PIMDP agent for some functionalities.

So in PIMDP, we already have, uh, really well written functions, um, for perception, um, plan, uh, and learning.

So the only thing we want to kind of replace is how the agent is doing planning and how the agent is taking decisions, uh, over policies.

Right.

So here, uh, here, um, we are using that parent agent class.

So from PIMDP dot agent, we are importing that agent class, um, which is sitting next to the SI agent that we are discussing now.

And, uh, basically we are taking in, uh, the generative model structure from the main program for this class to work, which is ABC and D and all the precisions and threshold parameter.

I mentioned, then, um, it is kind of, um, normalizing the prior preference that we mentioned in the main program.

So here, if I look at, uh, how C is, um, the structure of C is defined in terms of numbers and the prior preference is often interpreted, or it should be a probabilistic distribution for the computations to work.

Right.

So, um, we are going to normalize it as a probability distribution rather than, um, having, uh, having numbers that don't add up to one.

So that's what's happening here.

We are using softmax to do that.

Then what we are doing is we are initializing, um, the existing PIMDP agent, uh, with these, uh, generative model parameters.

And what we are, um, intending to do is, um, write a planning function for a given planning horizon and a given, um, threshold for three sets.

Okay.

So, um, there are three functions in this agent glass.

One is a helper function for planning, which we will discuss now.

Then there is a planning function itself, which is going to do planning using research.

Uh, and then because it is a recursive research, we are going to need a additional function that implements that recursive, uh, evaluation where, uh, we are going to call this function called forward search, uh, inside the function itself.

So we are calling this function inside this function.

So that is to calculate expected free energy for the next step.

And it will call it again for the next step.

Till we, till our planning horizon.

So that's the idea of recursive, uh, looping.

Uh, and finally, um, it will return the expected free energy, uh, for all actions, uh, given an observations.

And then we just implement the step function where it, it is written sequentially what to do given an observation.

Okay.

So, going back, uh, to the demo.

Here, uh, we have this first idea where you get an observation, uh, and it gives out an action.

So let's go to the agent dot step function and imagine what happens.

So if it is, uh, time t equal to zero or in the beginning of the experiment, uh, what it

is ideally supposed to do in the first place is infer that state using its observation.

So what we are giving it is an observation and using the modules for inference, it's going

to, uh, come up with a belief Q s, uh, which is an, which is a belief about states.

Okay.

So self dot Q s is the belief in inside the agent.
And once it has a belief about where it is right now, it can implement plan dot research,
which is, um, do planning for this particular belief of hidden state right now.
And once it has done planning, it can take decision using the sample action function in time DP,
uh, and basically return it, uh, return that action.
And if it, and for every other time steps, the sequence remains the same, but it is also
learning about the structure.
If you enable learning, uh, in, in, in, in your agent class.
So that's the step function, uh, but in order to do planning, what it does is it kind of reorganizes
the generative model structure for any number of hidden state modalities and any number of
observation modalities.
So to discuss how, uh, melting works, I would like to talk about, um, the new A matrices and
B matrices it, it evaluates for implementing that planning, uh, and let's understand that.
So let's go back to the original, um, hidden state factors.
So here we only have one hidden state factor.
Um, so that's why it is B zero, right?
And B one does not exist, uh, because we only have one hidden state factor, but imagine that
if I have two hidden state factors, um, with the same size, maybe, um, so here it, it could
be a location and maybe, uh, something else, uh, inside the agents mind.
Uh, and we should also have controls for these two hidden state factors, just like, um, so
there should be control for every hidden state factor.
Um, if you're familiar with, uh, active inference, uh, idea and then maybe the observation spaces
also directly observing these two hidden state factor.
Okay.
So this is a new generative model structure with multiple hidden states and multiple observation
modalities.
And right away, you can see that, um, the dimensionalities of your parameters change.
So this is a new model, you have, um, 25 observations coming from, um, 25 times 25 hidden states.
And if you look at the structure of the first observation modality, it's the same, but what
we want is a new matrix, uh, where it's going to be 25, uh, with, uh, not two hidden states,
but just one hidden state.
It's only a reorganization of, um, the generative model, but computations essentially remains the
same.
Uh, so that's what this helper function is trying to do, um, which will make things easier for
us when we have multiple hidden state modalities.
So, um, if you have multiple hidden state modalities, uh, we are going to compute how many total states
you have, which is the multiplication of number of hidden states in each modalities.
Okay.

So if you have 25 hidden states in one modality and 25 hidden states in the other modality,
you're going to have 625 total number of states.
And if you have four actions each in each modalities, then you have total of 16 actions,
which is nothing but the combination of these four acts each in, in, in the modalities.
So you, you're going to have four times four 16 actions.
If you have two modalities and it's basically, um, going to build a generative model, uh, that
has the same, um, model parameters, but just with a different dimension structure so that
it's easier for us to calculate the expected.
So now we have a new a and new B and a new belief, uh, which is nothing but, uh, tensor products.
So existing parameters and beliefs, it's nothing but a new big matrix, uh, and nothing else.
Okay.
It's not a transfer.
It's not a change.
It's just a transformation of structure.
And given this, uh, a B and Q, uh, we are going to predict what's going to happen in the future
and, uh, evaluate the expected free energies for them.
So in order to do planning, so that's the second function, what we are going to do is
first call the first function, which will do the melting for us and set up the generative
model in, in good dimensions, easy, uh, to compute.
Then we have the expected free energy itself for all the actions.
And then we have the probability, uh, that, uh, depends upon this expected free energy for
these actions.
So why is it just the actions and not the observations?
Because here we are going to, uh, evaluate expected free energy of actions, uh, for the
given observations, right?
So let me, um, go back to the slides and discuss, uh, this in a pictorially to make things easier.
So here we have the grid and we have the prior preference.
And what we are trying to implement is that if you observe some observation, uh, at time
t, then you're going to consider the consequences, um, of your actions, given that observation,
uh, because you can, uh, predict what's going to happen because in your generative model,
you have the transition dynamics that will tell you given this, uh, state, if I take this
action where I'm going to end up, right?
So that's basically predicting what's going to happen in the future.
And you're right now considering the consequence of, uh, uh, available actions in your arsenal.
And then if you take an action, then you can predict what's going to happen in the next time
step as a new observation, right?
So you have a probability distribution that tells you that say this observation is the most
likely, uh, and the other observations are not really likely.

Uh, then what you will do is you do this again.
You consider the consequence of doing your actions from that particular observation.
And, and this goes on in your planning depth, right?
So this can be thought of as maybe you want to go to the gym.
Then you are going to, um, consider all the consequences.
What will happen if I wear my shoes?
If I don't wear my shoes, uh, if I go in my car, if I don't go in my car, then you realize
that, okay, I have to wear my shoes.
Then you consider the consequence that I'm now ready to go to the gym and me going to
the gym will end up me being in the gym.
So that's the idea.
Like you consider the consequence of where you are right now, uh, and you can go as much
as you want, right?
You can predict.
So in a game of chess, you might be in a particular state.
You consider your consequences.
You see the future.
You consider consequences from that future, and you can go as deep as you want.
Um, um, um, you know, you know, you know, you know, you know, you know, you know,
you're right.
Um, depending upon your computational abilities, right?
So that's what you're trying to implement in this agent class, uh, where we are considering
consequences.
Uh, yeah.
So for every modality, we are going to consider the expected free energy for actions.
and this will basically call the next function which is forward search so forward search is
implementing the thing that i just mentioned considering consequences and in forward search
what you are basically doing is for every action so in line 149 i have a loop that goes over every
action then i'm going to consider the posterior or the consequences of all those actions i use my
transition probabilities to evaluate that consequences then i'm going to predict the
observations because my prior preferences are defined in terms of observations i'm going to
predict my observations and then evaluate the expected free energy which is the sum of risk and
ambiguity okay i hope that makes sense like here you have considered the considered the consequence
which is consequence of future which is post or posterior and you are basically evaluating how
good that posterior is depending upon your expected free energy and that becomes the expected free
energy for that particular action and you do this for all the action okay and why this is powerful
it is because you can go as deep as you want so here in the next step you go to this loop
where you will check if i am crossing my deep planning or the depth of planning and then you are doing

basically the same given that posterior what is the consequence of the actions of that particular posterior
so here for considering that we are again calling the parent function so the same function forward
search to consider consequences of those combinations and it will basically come back and add up your expected
free energy so what happens over this sequence is that you consider some or all future consequences
and then all that values will trickle up your tree and your that sum of the expected free energy will tell
you which action is good or which action is bad which you can take in to kind of see your preferred
observations so that's the idea of implementing tree search and i will also talk about the importance of
uh this threshold here uh which is which is which makes this uh algorithm possible so without this
threshold this algorithm uh will not work i will explicitly talk about why that is the case
and then once you evaluate the expected free energy for all available actions given the present state
you can basically compute um what you call the action distribution that is how probable is uh my action
or how i should take my action so that is um and we also have this action precision parameter alpha
so if alpha is very high then it basically is a highly skewed distribution where you will always choose the action
that minimizes expected free energy if alpha is really low then it's going to be a more um sparse or spread out
distribution and then you can use this action distribution uh to sample actions and so in in the agent environment loop
we just finished doing planning and computing that action distribution then using that action distribution you can
um sample an action from your policy space so let's look at the policy space in this generative model
so i'm switching back to the original generative model with one hidden state factor
uh and let's do planning and maybe um initialize this agent
i just want to initialize this agent uh to see the policy space not run the loop so
i initialize this agent say for planning depth of one and if i look at agent dot
uh
policies i can see that i have basically four available actions
uh which is north southeast and west and if i have an action distribution uh it will tell me
how probable is to take that action so if i look at agent
uh to pi so okay so this is not defined because i have not done planning uh but i can do planning and
then it will be defined
yeah so i implemented planning with research and then now i have an action distribution so for this
particular scenario i am going to take my third action the most which is 0.99 basically uh that's the
probability uh which is north south and east in this particular grade i just wanted to kind of
familiarize you with the matrices but we are now going to see um the agent environment loop in action
and yeah so now you can sample an action from um the sample action function and then implement learning
which is using the standard by mdp way where i will update my transition dynamics and likelihood dynamics
depending upon what i see and what's my belief and so on so my emphasis is on the decision making part

so once you sample an action then that action basically goes back um to the environment okay so now let us implement this for a planning depth of one and see how the agent behaves so here if it is a planning depth of one then that means that the agent is only considering the consequence of one time step ahead just seeing the immediate future for doing planning right so i'm giving the planning depth of one um i'm resetting the environment where the agent is going to start from that initial start state uh and and in the loop it's going to um get that observation take that action uh give back an action and we will look at the action probabilities and also we will uh give that action back to the environment get back the observation and this loop continues till the episode is terminated and i have set the episode length to be eight just to see the outcome of eight action okay so when we run this loop these matrices are nothing but the action distributions of how likely each action is to be taken and this is where the agent end up in the last time step okay so let's maybe kind of also enable this a environment dot render that will show us where the agent is at every time step so initially the agent was at this location and we have an action distribution of this here so north south east and west so the agent knows that it should go north why because if i look at the prior preference um this state is more preferred than this state right so the agent successfully calculated the expected free energy and inferred that okay i should go to this state uh and not stay in the state and because it has the generative model of transitions available it can infer that i should take that action north to go to this state so that's good uh and the agent goes to north and at this particular state the agent infers that it should go to north south east so it will take an action east uh and it will go here and at this point of time uh i want your attention where the action distribution is equally probable for north and east why is that because the agent is only looking at the immediate future right so let's go back to the prior preference where the agent is right now here or is it here yeah it is right now here in this particular state and if the agent is considering in immediate consequences of just one action then these two states are equally good for it to be in the next state so there is no distinction between these two states if it is only looking at the immediate future so that means that the expected free energies will conclude that i should go to north or east it doesn't matter if i am looking at just one time step ahead okay that's the idea and um out of probability it is going here uh it took the action east and from this um state when it's doing inference it's inferring that this state is better and basically it's end up ending up in this local maxima state which is this particular state where the neighboring states are less preferred and this is wall and you can you can't go there because it is forbidden for the agent by structure so it's basically going to sit there forever where it only sees that local maxima of prior to price and let's look at um what might happen if you have higher planning depth so if i go to the planning depth of three um then that means that the agent is actually uh reaching the gold state at the last time step uh but still in the third time point uh it had uh uh two probabilistic actions north and east so here uh from this particular state out of probability it took the action north um but it cannot equally take the action east and end up in this local maxima so let's run again and probably it will end up in this local maxima okay and only for

the planning depth of four which is sufficient enough uh which is necessary for this particular grid
um the agent is fully sure of what to do so at every time point it is fully sure of what to do
uh that it has to go north then east north north north east east and south and reach this
particular goal state so only for time step um or planning depth n equal to four it can successfully
navigate this grid um with 100 certainty so that's the idea that's the implementation that i hope you got
so there is also this idea of action precision so here it's a high action precision that is why it is
uh taking the actions that is uh from the probability if it is a low action precision um like one then the
good actions are more probable but that doesn't mean that um it will be taken right here uh by luck it is
taking the right actions and reaching the state but here probabilities are most fast but you will also see
like exploration behavior uh in more number of trials if you control this action precision so i set it to
a high value to make sure that the agent reaches the goal um for this particular problem but it's worth
playing and it's important right okay um yeah so for different planning um depths like one three and four in
this problem this is the behavior that you expect uh for lower planning depths which is not sufficient the
agent ends up in local maximas and or local minimas of expected free energy or local maximas of prior
preference but with sufficient planning depth it's able to navigate and reach the goal so that brings us to
the last point in this tutorial where why is it important to have a threshold uh in in uh evaluating sophisticated
inference so by threshold what we mean is that we can ignore uh future possibilities in two levels right
you can ignore um not likely actions or not likely observations in this research but if you consider the
consequences of all actions and observations that means that you have to consider four
consequences in the first place then you will have to consider four times the action states in
the next time step for the next time step then all of that multiplied with the number of actions
and this research becomes intractable until explode and it's even worse than the classical
active inference policy space problem but by defining a threshold of even a small value
will ignore possibilities so where is that implemented in the forward search algorithm um we are considering
actions uh with only action probabilities greater than the particular threshold uh here i am defining it as
1 by 16 also in the parent paper it's 1 by 16 um the action probability
probability okay uh so if it is 0 then that means that it will consider all the consequences of future
and that's interactable so you can ignore actions which is not probable and also ignore states uh which
is less likely or only consider states um that has probability greater than this particular threshold value
and that significantly reduces the computational complexity where you will only consider combinations that are
probably in in the future uh tree and that lets you go um deeper in your um planning horizon
that's an important point here and if you compare um the time that takes for deeper plannings
uh for a search threshold of zero uh so a threshold uh search threshold of zero zero means that you will
consider all consequences and the more deep you plan um the more time it takes and if you see uh to consider
only the first future or the immediate future it takes 0.01 seconds for considering three possibilities into the
future it takes three seconds and for four it takes 300 seconds then you can see that the computational time

is
exponentially growing uh but if you have a very small threshold threshold um you have that computational time uh that makes sense uh for implementation uh in in real world right so here for n equal to four that is four that is four times steps into the future it is only taking 0.1 seconds and that's okay it i can still do simulations with this complexity but there is no way i can talk about how this is um how less the computational complexity is it truly depends on uh the nature of your prior preferences and environment in action but this search threshold actually works in in real life and we just saw that in our simulations right uh for n equal to four it took only like 0.3 seconds in our simulations but if you set the search threshold as zero um it already is 300 seconds for doing full depth planning and if i set a planning depth of five then it will basically run forever maybe and i i will not be able to do uh symbols so that's the idea of search threshold um so actually that's it uh i wanted to explain the agent class the environment class and the particular demo uh and yeah maybe it's a good time for questions if if anybody was listening uh and i hope people um get to play with this code um and look at the tutorial and and implement this and uh build generative models like this so this particular example is how do you build a generative model for this gradual task and see how the agent is able to take meaningful uh actions but here we gave it the true structure of the environment in the b matrix and a matrix and so on but um you can also play around with learning uh in in the sense that while defining uh the agent dot step you can add a flag uh that says learning equal to true and if you start with an uninformed a b and so on you can experiment on how the agent is learning that environment here you can look at the b matrix in the beginning you can look at the b matrix after say 10 trials and see how the learning is taking place uh here it doesn't matter because the agent knows the structure and it won't learn much but if it starts from a unknown structure then there is scope of learning also to be implemented and it's already implemented because we are using uh existing pymdp functionalities
uh for learning a and b and it's already part of the step function so i hope step function is clear
um which is the only thing you need to know if you're trying to implement sophisticated inference
and just the names of these matrices if you want to probe them and look at them
uh and yeah i hope the session was uh useful so i thank um my collaborators and uh connor who maintains pymdp and also brandon who runs the pymdp fellowship which i was part of and that's where i worked on implementing sophisticated inference in pymdp and it will be part of um the original pymdp module soon and i hope um people can start using this module to simulate sophisticated inference experiments and this basically becomes useful so maybe it's a good time to discuss questions or clarifications on the code or maybe it's a good time to take a break as an able to switch great thank you that was awesome
well i have a few different questions and i'll read a few from the live chat so i'll first just go to the live chat then um ask those and then ask some other questions so dave asks how do you think about the neural implementation of recursion brains don't seem to implement computer hardware style recursion deeper than a stack depth of one aside from heavily over learned tasks we can confine ourselves to asking about recursion for the purposes of exploring temperately deep state spaces searching forward in time so how do we how do we reconcile this really beautiful and elegant and computationally

efficient full depth research with the biological basis of multi-scale planning yeah
um so i'm not an expert in neural computation uh but the answer to that would be um
basically you're doing only one computation at a time right and you have or all you need is some memory
to store your beliefs and use those beliefs to kind of do the same computation again so um we are not
talking about uh newer i mean this hardcore recursive implementation we are only doing local computations
at a time and just because of the structure of the generative model and we and because we have memory
um this this can be done and and i don't see why uh brain can't do it even though individual neurons
might not be able to do it the main brain has memory the brain has the ability to store memory and the
ability to dream the ability to simulate uh it knows the consequences of actions and you do this on a
daily basis um where um you plan your future and decide what to do right so on a single neuron level i'm
not really sure uh of how to answer that question but i don't really see why the brain can't do it as an organism
cool um okay to the code i guess i have a few questions let's can we go back to the maze
and of course if anyone else has questions in the live chat just go for it um so in in the maze how how
does the uh the moves that are possible how is that reflected what what stage is it updated that what
for example that it initially knows that it can only go up and then it can only go right or down
where is that reflected with the updating of what what the inv like kind of that relational aspect of
affordances what is even possible to do and then how does it evaluate in a deep tree search
when it does does it need to know what things could or couldn't happen in the future
um so if i understand the questions correctly um there is import there is an important distinction
between the generative process and the generative model right so in the grid world um which is the
the generative process um we have implemented all of that manually where we have this transition dynamics
um that already knows what will happen if you take an action from a state so it's like an environment
that knows how to work um so it's like a reality where there are consequences of actions and it's already
there but this information is available to the agent uh to be part of its generative model and in the
generative model what it basically does is use that transition dynamics which is given or learned to
simulate what might happen in the future okay so once i have that transition dynamics um if i go to the
the research what it is essentially doing is um evaluating the posterior um given a particular state
and an action j so from my generative model i know that if i start from this state and if i take this
action i go to this posterior and i consider all the consequences and if in the generative model it is
unlikely that if i go east i don't go there and so on it's basically it's it's automatically reflected in
the expected free energy um so i hope that answers the question so here um if i set the action position to be
high and also enable the where might um so initially i am in this particular state uh and we with respect
to the expected free energy what i am doing is i am using my generative model to predict what will
happen if i take four actions and my predictions will say that if i take north i will go here and if i
take all the other actions i'll stay here and because in my prior preference north is more preferred i can
infer that north is the uh action i should go for so that's the idea so here the grid structure is given
to the agent and it might be a little confusing but the agent can also learn this grid structure and

if this will work so once i know the grid structure as an agent i can simulate the future and consider the consequences of action right so that's what's happening and once i'm in this state i will
consider the consequence of all four actions and i will infer that okay um going east is better because that will take me to this state so there is always a difference between or you should keep in mind how generative model and generative process are two different things uh and what the agent knows might not be the reality or might be the reality depending upon what you give it hmm cool so if you were gonna go about making a new situation that you wanted to do generative modeling for do you tend to start from an existing working py mdp notebook and start modifying state spaces or do you draw it out on a canvas like how would you recommend somebody other than replicating what is shown here let's just say we're interested
in something that's not exactly just a maze what do we do to get our head around how we should proceed yeah um good question so if you're trying to simulate say a new environment uh you have the heavy lifting uh to do which is to define uh the generative model for the agent you can either define a very sparse generative model which the agent can learn but you have to define the structure that structure should be there uh and only using the structure the agent uh can learn the generative model so here you can make use of this cell of code to understand how i define the structure of the grid
okay so i am defining a structure for the agent for it to make use of i am saying that there are 25 valid states uh and there are four available actions and um this is the standard way of defining the state space in py mdp and you also have to define um the central parameters a b c and d for the agent simulations so here i am defining a using my state space and observation space but this step of giving it or telling it that it's an identity matrix is my decision choice in my modeling i don't have to do this for simulations i can see if the agent is learning it from a random a matrix or when it starts from a random a matrix and similarly for this b matrix this structure is defined and there are functions like this which can give you a random b matrix but this is a modeling choice where if i want to give it the grid structure or not give it the grid structure i can start from a random b matrix let the agent learn and look at that learned b matrix just for the purpose of the demo i gave it the grid structure to enable it to take the actions but it's not a mandatory thing so this notebook is useful in the sense that you know what to do but definitely you should play around with steps that may not be mandatory right so if i give a prior preference for this state to be the maximum then you can see behavior that the agent will try and go there right so um this prior preference is defined uh in conjunction that this is the goal state but this may not be the goal state and in a different task what prior preference means is different according to the environment um right so that's also there and your prior so once you define that generative model which you have to do that's you can't run from it then everything is kind of automated the agent um only have to use the agent dot step give it the observation from the environment the agent knows how to take actions and then um everything that happens inside the agent you don't really have to worry um yeah so this structure i'm sure will be useful for your particular task that you're trying to model in your
um

yes very interesting and how would you contrast that or or point to any similarities or differences with
how this would be pursued outside of active inference like if somebody were just going to use another kind
of deep policy agent in the maze example um what parts of the process would be familiar and what parts
would be like a lot of work that wasn't expected or skipping through parts that were a lot of work
otherwise yeah um so the general structure is very familiar to somebody who does things like this uh in
green personal learning so um the idea that it's an agent dot step and environment dot step so this is
the standard open ai gym way of writing an environment and a standard open ai way of writing an agent
okay so if you have a q learning uh agent who does the same trying to navigate then the way you have to
define uh the q matrix is the heavy lifting there it's just a state action mapping and uh in contrast to
that in active inference you have to come up with a generative model that you want to see so in active
inference it generative model is the central thing right without that without a generative model there
is no meaning of purposeful behavior in active inference so the only unfamiliar part for a person
who is coming from say a field like reinforcement learning is the structure of the generative model
but there is no way other than getting used to it where it's the pomdp structure which dominates
but if you're doing deep active inference all of this is going to be neural networks and pomdps are
also not active inference um things right it's a industrial engineering thing uh so pomdps must also
be familiar for people who are coming from the computer science background just the idea that um what
really happens in the agent is uh the active inference part where we have expected free energies and
variational free energies and if you want to learn about that then you have to go to the agents and
see how it works look at the matrices numerically see what's happening but i don't in a level where
you want to get started i don't see any problem all of this is um standard um frameworks like pomdps and
open ai gym environments agent environment loop all this is very uh deeply discussed in computer science it's not
um so um so what other um motifs or cognitive phenomena are you excited or how do you see the pi mdp
development trajectory continuing after your sophisticated inference gets pulled in yeah um so the
pi mdp had the original functionality and um the functionality to implement or simulate um general
active inference agents with the policy space and so on and that enabled a lot of people in the community
who um is not who are not um familiar with um complicated coding and so on who people who do psychology
uh psychiatry and and all the things right so whoever want to come up uh and try implement active inference
by mdp enabled that and i'm i'm hoping that this module will enable people who want to try out
sophisticated inference experiments in their particular domain so this if you uh spend some time and get
familiarize yourself with the structure of how pi mdp works then everything else is just writing a
jupiter notebook with minimal code right to simulate this so if you have a particular task in your domain
i don't see a problem for a beginner to kind of try and code it and what i'm very excited to see is
people using this module for variety of experiments just like how people started using pi mdp and sophisticated
inference is taking off uh and it's now uh widely talked about about how it is um how it is the way of
doing active inference and i'm really hoping that people in various domains start using this module and

see their experiments and i look forward for the feedback yeah so what pi mdp did two years ago i'm
hoping this module will do to people who are trying to uh model active inference in the state
so you mentioned the open ai gym and the standardized format and and what benchmarks do you use or what
are you what kind of test suites are you comparing and how do we really know when we've made a generative
model that really exceeds or excels in a way that other um techniques are just not doing
yeah so if i um may go to the open ai gym website um we have several experiments there the classical
reinforcement learning examples um like the lunar lander that you see in this uh screen right now
so active inference from its inception has faced problems of scaling um to tasks and that has that's
in itself a field of research in active inference um scaling active inference and that's one of the reasons
why deep active inference took over um dealing with tasks like this so there are benchmarks um even now
where the sophisticated inference may not be able to deal with state spaces and personally that's my
research in my phd i am actually looking at um optimal optimizing computations in sophisticated inference
algorithms that lets you scale up to environments like that but to get started
um you will have to kind of write code and see if it works for an environment then look at um if it's not
working then you have to look at methods to scale it up and and so on so if i am talking about benchmarks
sophisticated inferences as good as any rl algorithms um for this state space so for small problems
so which gate inference will work and it's really good but for high dimensional problems like this um the
classical implementation that i just showed might not work but it's good enough for any um decent
experiment but if you want to scale up then that's still open and it's a new field of research and what
you do might become a next new important paper right so that's all i can tell in that regard cool you have to
work and see yeah what measures do you think you'd be looking for like uh computational resources or
what what are the measures that even make sense to juxtapose such different methods
yeah so the openai gym was uh designed for that to compare different uh algorithms so openai gym uh by definition is a collection of many environments
so in my demo i was talking about the grid environment openai gym is nothing but a collection of many environments
which will let you interact with those environments using the environment dot step function
um yeah so here we have the environment dot step function that will let you interact with the lunar lander
and that particular task will have matrices that lets you um judge how good or bad your algorithm is
so in uh this lunar lander problem how optimally can you land your rover between these two flags
by spending minimizing the fuel and so on so those matrices are very tasks specific and that's one
direction you have to take you can take uh try and compete uh with rl algorithms in in matrixes but um
the right potential or the potential i see in sophisticated inferences uh modeling intelligent
behavior where in rl the focus is to get things done to make this work but it's not really explainable
especially deep rl and deep learning methods but in active inference if you manage to scale it up they
are explainable and and they are um that will let you understand how intelligent emerges with time

and i i i see that more interesting than um competing with rl because if your focus is
getting things done then maybe engineering is the right way and not active in practice
any other comments or thoughts
aswin do you have any other comments or thoughts no i i'm pretty happy i hope um i was clear explaining
the code maybe it was too complicated or symbol depending upon your level um but i hope it is
useful to at least one person who would uh start using this and write the code and so thank you so much
for your time awesome thank you for the opportunity thank you for joining till next time see you
thank you so much bye
all right
greetings takuya hi until next time see you
you can um mute the live stream or turn off the other live stream but yeah thank you for joining
yes uh thank you for uh inviting thank you for it yeah
um well
we have a a little bit of 35 or 38 minutes so it would be awesome to uh have your presentation
um
uh
uh
the moment uh
well
can you see my well i saw a powerpoint
uh
uh
can you see my screen uh perfect perfect okay
so shall we start yes thank you okay oh then uh thank you for uh uh organizing this wonderful uh
symposium today i'd like to talk about uh
uh
relationship between canonical neural network and uh
active inference uh and uh possible extension uh for modeling uh social on the shared intelligence uh
using canonical neural network so let's start
um
for surprise minimization and by this process organism can perform
variational BGN inference of external millennial state and this is this cartoon just show one
example typical setup under the free energy principle active inference here there is a
hidden state in the external world and only a part of this state can be observable for our agent this
doc and this transformation is done by a genetic model parameterized by theta and to infer the
hidden state agent need to reconstruct the copy of the external state called the posterior belief
and this optimization is done by minimizing variational free energy and parameter is also
optimized by minimizing free energy to to obtain an apt generative model to represent this relationship

an interesting aspect of the free energy principle active influence is its application to the
optimization of action here our agent have some preference prior C and to obtain this observation
in the future agent may select the action that minimize the expected free energy in the future
to to to to obtain the most predictable outcome and finally by selecting action that minimize the expect
frequency it can obtain the feed so this is a typical setup under active influence but question
is what is the neuronal substrate that can implement this process so that that is our interest
and to address this issue we propose a theory as follows here we consider that external world is
parameterized characterized by a set of the variables like this here this is a posterior expectation and
the s indicates the hidden state index on world delta indicates the action of the agent
the decision and the theta indicator parameter and the lambda is a hyper parameter so those set of parameters or variables characterize a generative model and free energy
V is defined as a function of the sequence of observation and external states

and its minimization indicate the variational free energy, variational Bayesian inference.
Similarly, we consider dynamics of neural network and we consider that dynamics is characterized by
a set of those variables. Here x indicates the neural activity firing rate at the middle layer
neurons and y indicates the neural activity of output layer neurons and w here is a synaptic
weight and phi is any other free parameter that characterize the neural network. And we consider
that neural network dynamics is characterized by minimization of some cost function l. This is a
function of the O and the phi internal state of neural network and its minimization indicate the
generation of neural network dynamics including both activity and this plasticity. And our theory
indicates equivalence between those two functions meaning that for any neural network that minimize
some cost function l there is a generative model that satisfy if we call l which means that
the recapturation of external dynamics is an inherent feature of any neural system that follow such
dynamics. So this is an interesting prediction but too abstract. So I'd like to introduce
some analytically tractable example to understand this relationship formally.
So first we consider a very simple architecture. This is represented by a POMDP without any state transition. So
hidden state s is simply generated by prior distribution d and it's a it is a binary
state but we consider a vector of binary state. So it's a factor of factor of structure.
And observation is also a binary vector and the transformation from S to O is characterized by
categorical distribution using likelihood matrix A. And variational
Bayesian inference indicates the inversion of this generative process like this. So by solving the appropriate free energy
functional we obtain those posterior brief that is the optimal in the Bayesian sense. So on the other hand for

neural network we consider such structure.

Here upper part indicates the generation of signals in the external world.

And the neural network comprises only single layer. Here activity of output layer are generated by a weighted sum of

sensory input O weighted by synaptic matrix W. And we then characterize this neural network

in the next slide. So to model neural network or neuron. We start from considering Poziken-Hacksel equation which is

four which comprises four differential equations. And this is a very complicated non-linear equation.

So we consider some reduction of the equation. And also it is a very plausible. So we consider some reduction of these equations.

So a typical reduction method is like this. For example, M here is much faster than other variables. So this can be replaced with its fixed point.

So it is known that H and minus one, sorry. It is known that H and one minus S, N is, have a similar dynamics. So we can consider a new effective variable U to characterize those two variables.

And then we obtain 2D Hoziken-Hacksel equation like this. So this is a famous class of neural network model which includes the well-known Fitzpieu-Nagumo model

or other continuous neural network model. And we modify those model to derive canonical neural model. So this is a definition of canonical neural network model.

Here, leak current is characterized by inverse sigmoid function instead of cubic function which is adopted in the Fitzpieu-Nagumo model.

And we also consider a connection, synaptic connection. And this part indicates a synaptic input from a sensory layer through weight matrices.

And one may consider that W1 indicates excitatory synapse and W0 indicates inhibitory synapse.

And the threshold are adaptive thresholds that are function of W1 and W0.

Interestingly, when we consider a fixed point of this differential equation, we obtain a well-known rate coding model

with the sigmoid activation function. So, which means that we can say that in some sense,

this canonical neural network model is an approximation of Hoziken-Hacksel equation and its approximation level is, in some sense,

between the realistic model and the most simplified rate coding model.

So, we basically consider this type of neural network model. And in the next slide, we consider what is the plausible cost function for this neural network model.

So, again, we consider the same equation for canonical neural network model, which represents the activity of neurons, vector of neurons.

And we consider the cost function for the differential equation, which can be obtained by simply calculating the integral of the right-hand side of this equation,

and get this type of cost function for neural network model.

This is a biological plausible cost function in the sense that its derivative drives neural network activity, which has a certain biological plausibility.

Moreover, if we consider the derivative of this cost function with respect to synaptic weights W, we obtain a conventional synaptic plasticity rule, which follows a Hebbian law.

On the other hand, for Bayesian inference, we first define generative model like the previous slide,

and we then derive the, sorry, we then derive a variational free energy for a given generative model.

So this variational free energy is derived from the POMDP model in the previous slide,

and its minimization indicates the Bayesian inference and the learning.

So we found a formal correspondence between components of those two cost functions,

like this.

So this block vector formally corresponds to this block vector that represents a posterior expectation,

and this logarithm corresponds to this logarithm.

And actually, this A matrix can be represented as the block matrix like this,

and its dot product corresponds to this computation.

And finally, this phi naturally corresponds to this log D, log of state prior.

So, which means that because the cost function are same, its derivative provides the sum, sorry.

So, because the cost function are formally equivalent, its derivative, its, sorry, its result of derivative also corresponding each other.

So, which means that for any neural activity equation in this form, there is a corresponding Bayesian inference equation.

This is a equation that compute the posterior belief of the hidden state.

And this synaptic plasticity equation formally correspond to learning or parameter of generative model.

And moreover, by establishing this relationship, we can consider the reverse engineering of the generative model from empirical data.

Here, this schematic summarizes our approach to reverse engineering of generative model.

And we first record the neural activity and assign the canonical neural network to explain this obtained data.

And by computing the integral, we obtain the cost function for this canonical network.

And by the mathematical equivalence we established, we can automatically identify generative model and variational free energy that correspond to this neural network architecture.

So, interestingly, this is a Bayesian agent, which is a type of, which is a kind of artificial intelligence.

But importantly, this artificial intelligence is formally derived from empirical data.

So, we can say that this agent is a biomimetic artificial intelligence that follows the free energy principle.

And then its derivative with respect to parameter posterior derive synaptic plasticity algorithm that follows the free energy minimization.

And its time integral can predict the learning process of original neural network data.

So, which means that if the free energy principle is correct, then this prediction should work, should work, should work, and should be able to predict the result of this neural data without referencing to the data itself.

So, our strategy is, in summary, our strategy is that we reconstruct the generative model only from the initial data, like initial data before learning, and then predict the learning process or learning curve that this neural system should follow.

So, we're going to compare, and then predict the data by using the free energy principle.

And compare or examine whether this prediction is correct by comparing actual data after learning and the prediction by the free energy principle.

And if this prediction is correct, then it indicates the predictive validity of the free energy principle under setup considered.

Okay.

So, we apply this strategy to the in vitro neural network.

So, here, this in vitro neural network are stimulated using the POMDP generative process defined in the previous slide.

So, there are two hidden sources that are binary signals, and they are mixed to generate 32 sensory stimuli.

So, there are two hidden sources that are binary, and those are binary.

And this is an overview of the experiment.

By stimulating in vitro neurons, they generate spiking response, and those lines indicate a high density spiking response.

So, we observe that in some neurons, the response specificity is, so we found that for some neurons, response was high to source one signal compared to source two signal.

And if we see the transition of those neurons, we found that although we remove the offset at the first session to set those activity zero, but we see that those neurons self-organized to response high to the

high when source one is one, but those neurons response low, low level when the source one is zero.

So, which means that those neurons activity, those neurons response was consistent with our theoretical prediction that neural activity self-organized to encode the posterior expectation of hidden state.

And we also found that the neural activity, and we found that the neural activity is encoded by neural activity.

And the next question is, what about the neural substrate?

And we then asked whether, we then asked whether, we then asked if the prior belief about hidden state is equal to, effectively equal to the firing of the neural activity.

model, neural activity model, neural activity model, neural activity model, neural network model.

So, if the theory is correct, this is correct.

correspondence correspondence correspondence should exist.

To check this, we first simulated a Bayesian agent.

and when we varied the prior belief of the Bayesian agent, the inference was attenuated as expected.

And we found that when we varied the excitatory level of in vitro neural network by using the pharmacological manipulation,

we also found that the attenuation, the attenuation of inference.

So, which is consistent with our theoretical prediction that firing threshold encode prior belief of the hidden state.

And next, we consider whether the cyanatric plasticity follow the free energy principle by asking whether free energy principle can predict

the qualitative self-organization of subsequent neural data.

So, here we model neural network like this.

There are two ensemble neurons that encode source one and source two.

And we first compute the effective synaptic weight of those networks using a conventional connection strength estimation approach.

And the plot those synaptic weights on the landscape of theoretically computed free energy.

So, this is a trajectory of empirically estimated synaptic weights, effective synaptic connectivity.

And as predicted, those changes reduce the free energy.

And here we computed this theoretically predicted free energy landscape only using the first 10 sessions data.

So, that this indicates some prediction of the self-organization.

And for more explicit prediction, we then simulated neural activity and plasticity using free energy principle.

Here, the brighter color indicates the prediction of data without reference to activity data.

So, those lines exactly follow this free energy gradient.

And we found that this predicted trajectory is tightly correlated with this empirically estimated effective synaptic weights.

And the error rate is like this.

So, it indicates that free energy principle can quantitatively predict the self-organization in this setup.

So, it indicates some predictive validity of the free energy principle under this setup.

And then, we also consider the modeling of the neural modulation using active inference.

It is well known that synaptic process is modified by various factors like dopamine, noradrenaline, acetyl, gorein, serotonin, and so on.

And one interesting property of those modulation is that even though dopamine was added after associative plasticity was established, it can change the result of plasticity in a post hoc or retrospective manner.

So, it implies some association to the reward and past decisions.

So, we model this process using canonical neural network.

So, here, we model the post hoc modulation of heavy plasticity using this type of plasticity equation.

And we also consider the recurrent neural network structure and output layer for this network.

And we consider that modulation occurred at this connectivity layer.

And then, we found cost functions that can derive those differential equations.

And then, found corresponding variational free energy and generative model.

So, which means that this sort of neural network activity, including modulation of synaptic plasticity,

can exactly follow the free energy principle and some type of home-DP generative model.

And by using this, we show that this biological plausible neural network model with modulation of heavy plasticity can solve some sort of delayed reward task, like a maze task.

And then, finally, we would like to discuss a possible extension to this framework, to the modeling social intelligence.

So, to infer our conspecifics, we need to select an appropriate generative model for our partners, depending on our partners.

So, this can be done by the Bayesian model selection scheme.

And we previously proposed a model that can predict the multiple barathons using one big generative model that comprises multiple generative models.

And this movie shows a prediction of songs.

So, like this, this model nicely identifies which generative model is the best to explain a given sensory input.

And this process indicates that the model can correctly input the appropriate model.

And then, imitate the song by its own action.

So, although in the previous work we didn't discuss a detailed neural substrate for this mixture generative model, we now be able to consider the corresponding circuit architecture.

For example, if we consider the modulation of module, neural module by neuromodulator like dopamine, it acts as an attentional filter.

And this attentional filter can be explained by three-factor heavy and running rule introduced in the previous slide.

So, again, this modulation works as the post hoc modulation of past heavy plasticity.

And this modulation can optimize each model to represent one generative model, one generative model, one song in a mutually independent manner.

So, through this process, it is possible to learn multiple generative model in a virtually plausible manner.

So, in summary, we found that the dynamics of canonical neural networks that minimize the cost function can be read as a minimization of variation of free energy.

So, in summary, it indicates a principle is an apt explanation for this type of neural network.

And we also validate this prediction using some in vitro set trap by showing that free energy principle can quantitatively predict the self-organization of subsequent plasticity,

only using the initial data.

And as the modeling, we can extend those canonical network modeling to the action generation and planning via the delayed modulation of heavy and plasticity.

And finally, we discuss a possibility to extend this canonical model to modeling the social or shared intelligence to interact with multiple partners.

So, that's all of my talks.

Thank you for listening.

This is an acknowledgement.

Our collaborator and fundings and our unit are now recruiting postdoc researchers.

So, if you're interested in it, please check.

Thank you for listening.

Awesome.

Thank you for the presentation, Takuya.

I'll just ask a few quick questions from the live chat and a few other things that come up.

So, Dave says, Takuya used the phrase after plasticity was established.

Was his group able to modify, e.g. increase plasticity, or is he saying merely that it was shown that there is plasticity?

So, yeah, I'm not sure if I understand correctly your question.

But those groups showed that dopamine, adding dopamine after two seconds after the association occurred, can change the magnitude of plasticity.

So, without, how to say, if dopamine addition was before this association, then plasticity level is low.

But after dopamine addition, after association can change this level, can increase the plasticity like this or like this.

So, which indicates the post hoc modulation.

Cool. I think that answers it.

Well, what are you excited for?

Or what are your hopes or feelings on where the active inference ecosystem is at and where we're headed in the coming months and years?

Well, sorry.

Sorry.

Again, please.

It's a...

Just what are you excited about?

Other than your own research directions, what are you excited about in the active inference ecosystem?

Hmm.

Yeah.

So, of course, one direction is the modeling of social interaction.

So, which is much a rich architecture than the interaction between the static environment.

So, if both agents run with each other, then many interesting phenomena can be observed.

So, we are excited with modeling those phenomena using biologically plausible network, neural network model through this equivalence.

Awesome.

Any last comments?

Any other comments that you want to make, Takuya?

Okay.

Okay.

Okay.

Okay.

Awesome.

Thanks again for the presentation.

And, uh, people should check out Livestream 51, where you and I talked a few other times and went into some of the details on that work.

There's a really... there's a lot there.

It's really exciting.

Thank you.

All right.

Thank you.

See you next time.

All right.

See you next time.

Bye.

See you next time.

Bye.

Bye.

All right.

All right.

All right.

The next session is going to be with Shannon Dobson.

This section is going to be called Dark Imaginarium.

Shared Intelligence as an Infinity Curiosity Type.

Um, I'll message Shannon, make sure that everything's good with the audio.

Wow.

Great talk so far.

Welcome, Shana.

How are you doing?

Yeah, I'm doing good.

Um, awesome.

Well, please take it away.

Thank you for joining.

Awesome.

Let me see my screen.

Okay.

Yep.

Looks perfect.

Thank you.

Okay, perfect.

Hey, everyone.

Thank you so much for the invitation.

It's such a great honor to be here.

I'm learning so much.

I'll be talking about, um, an idea I had called Dark Imaginarium, where I'm trying to make shared intelligence the concept of a, what I call an Infinity Curiosity Type.

So you were so gracious to link the paper so anyone can find more details there on Phil Archive.

So I want to start with an ontological, a discussion about ontological commitment.

So my first question is might we re-examine our ontological commitments?

So in particular, two of them.

One, um, Sean Carroll said so amazingly, but cannot be known because it does not exist.

So Sean Carroll says the Heisenberg uncertainty principle is often explained as saying that we cannot simultaneously know both the position and the velocity of any object.

But the reality is deeper than that.

It's not that we can't know position and momentum.

It's that they don't even exist at the same time.

The second one I want to examine is Lenny Suskind's quote.

We are all behind someone else's event horizon.

So Suskind has a brilliant idea of space time emergence as the entanglement of two black holes.

And what emerges from that is called computational complexity.

So the world is quantum.

So every theory should be derivable strictly from the laws of quantum mechanics.

One is going to give us the notion of shared.

Two is going to give us the notion of ecology.

So what would an ontological commitment to dark look like?

So my continual goal, as everybody knows, is to sustain a simultaneous experience and to understand why I currently cannot.

To sustain a simultaneous experience would be to bypass the integrative system we call consciousness.

So how would you attain a two consciousness in the sense of in category theory or an in consciousness?

How would we actually develop something called like a two memory?

So my colleague Robert Pretner and I fleshed out something called an awareness.

We're sort of examining this idea.

Mike Levin and all have repeatedly shown that planaria memory is not in the head.

Where is the memory?

Is memory a sustained superposition of what?

And also, what is the structure of an amnesis?

Could you have a superposition between memory and an amnesis?

So I had a small dialogue with Chris Fields, the amazing Chris Fields.

And I said, you know, what is the difference between these two questions?

What is the waking state?

And what is consciousness?

Or could it be that these are entirely separate questions?

So we're sort of in a mess either way.

If there is a difference in these questions, then it is possible to sustain a waking state without consciousness.

And or a conscious state in a non waking state.

So imagine somehow being in a waking state under general anesthesia.

Well, if these are not entirely different questions, then we could contend that consciousness is a form of waking state.

Or something along the lines of a waking state is a condition of a conscious state.

So how can humans have delta waves, which are canonically indicative of unconscious deep sleep, and that's a non waking state in the REM state, the patterns of which are indistinguishable from a waking state.

So we all know that general anesthesia, something like a propofol, can reduce the polyrhythmic brain activity to one uniform hum.

And we know there's as many theories about what anesthesia does to consciousness as there are anesthesiologists.

But we've repeatedly posited consciousness as the proto state.

And I wanted to turn this idea on its head, like, is consciousness so robust or is it a delicate interference or is it a delicate inference?

Now, we know beginnings are always tricky, but what if sleep was the proto state?

So if, you know, if life is oceanic and it revolves in these deep pressures, with the pressure of getting onto land, like squeeze the REM state into this full blown waking state that we're engaging in right now, anybody who's listening.

So then waking state would be something like an acceptation, which is, I think, quite impressive.

So it's in this mess that I attempt to define dark consciousness as a mixed frequency state of being self aware or conscious while simultaneously being in a deep sleep, specifically in three sleep.

So the sleep is proto state is quite is quite strange.

The REM brain activity is indistinguishable from waking activity.

And this sort of enchanted it's enchanted me for a bit.

The only difference is eye movements.

So either blinking versus lateral movement.

So REM is naturally having mixed frequency brainwave states, but the N-REN states and the REM states are starting to blur as delta waves are creeping into the REM state.

Now, they've also found that sleep is a local phenomenon.

So of course, I'm going to ask the Shanna questions.

So then who is sleeping?

Who is remembering?

Who is dreaming?

Lucid dreaming is a quantum superposition or is it a superposition or an integration?

So in a sleep state, you know, can then can we sustain a simultaneous experience?

So must the eye be knocked out in order to sustain this level of simultaneity?

Dreaming, we know, is characteristic of unsynchronized brain activity.

So is the question then, hmm, my eye cannot sustain a multi-consciousness, but my dark eye, my sleep eye can?

You can reframe this in terms of reporting mechanisms and say, perhaps we are conscious of things we don't think we are conscious of.

So let's talk about what is dark and a canonical definition of a dark theory or anything like this is like, you know, dark matter does not interact electromagnetically with matter.

But I want to sort of play around with another idea of dark coming from Deleuze's logic of sense, something he calls the paradox of infinite becoming.

And which I'm going to read on the next slide. So what if dark was the simultaneity of a becoming that eludes the present?

Let's hold on to that for a second. Is dark a resolution issue?

But resolution is merely translation breaking translational symmetry.

So is dark a symmetry breaking issue?

And if so, what symmetry and are we actually looking at the wrong symmetry?

Is dark a sustained quantum superposition? But of what again?

Is dark a many worlds branch? Is dark the dolphin uni hemispheric sleep mechanism?

So I'll read from Deleuze's logic of sense.

Alice and through the looking glass involve a category of very special things, events, pure events.

When I say Alice becomes larger, I mean that she becomes larger than she was.

By the same token, however, she becomes smaller than she is now.

Certainly she is not bigger and smaller at the same time. She is larger now. She was smaller before.

But it is at the same moment that one becomes larger than one was and smaller than one becomes.

This is the simultaneity of a becoming whose characteristic is to elude the present.

Insofar as it eludes the present, becoming does not tolerate the separation or the distinction of before, after, or of past and future.

It pertains to the essence of becoming. To move and pull in both directions at once.

Alice does not grow without shrinking and vice versa.

So good sense affirms that in all things there is this determinable sense or direction.

A paradox is the affirmation of both senses or directions at the same time.

I continue.

The paradox of this pure becoming with its capacity to elude the present is called the paradox of infinite identity.

The infinite identity of both directions or senses at the same time.

A future and past.

Of the day before and the day after.

Of more and less.

Of too much and not enough.

Of active and passive and of cause and effect.

It is language which fixes the limit.

The moment, for example, at which the excess begins.

But it is language as well, which transcends the limits.

And restores them to the infinite equivalence of an unlimited becoming.

Hence the reversals which constitute Alice's adventures.

The reversal becoming larger, becoming smaller.

Which way? Which way?

Asks Alice.

Sensing that it is always in both directions at the same time.

So that for once she stays the same through an optical illusion.

The reversal of the day before and the day after.

The present always being eluded.

Jam tomorrow and jam yesterday.

But never jam today.

The reversal of more and less.

Five nights are five times hotter than a single one.

But they must be five times as cold for the same reason.

The reversal of active and passive.

Do cats eat bats?

Is as good as do bats eat cats?

The reversal of cause and effect.

To be punished before having committed a fault.

To cry before having pricked oneself.

To serve looking glass cake before having divided up the servings.

All these reversals as they appear in infinite identity of one consequence.

The contesting of Alice's personal identity and the loss of a proper name.

The loss of the proper name is the adventure which is repeated throughout Alice's adventures.

Hence infinite identity.

So I take this and I'm going to formally define dark consciousness as a state of being conscious.

While simultaneously being in deep sleep.

Specifically in three.

Where dark is going to be a two-fold hybrid.

The simultaneity of infinite identity.

And of a becoming that eludes the present.

So let's reconstruct the dark consciousness version of Deleuze's paradox of pure becoming.

For Alice in dark wonderland.

And through the dark looking glass.

Okay.

When I say Alice becomes sleepy.

I mean that she becomes sleepier than she was.

By the same token however.

She becomes more self-aware than she is now.

Certainly she is not asleep and self-aware at the same time.

She is asleep now.

She was self-aware before.

But it is at the same moment that one becomes sleepier than one was.

Or more self-aware than one becomes.

So dark consciousness I contend would take place in something called like a dark time.

This is going to get weird.

So Chris Fields and I talked about an entropic time.

Where a time where learning exceeds forgetting.

So we all know that linear time is based on the linear ordering property of the positive integers.

Okay.

So what would be the structure of a time that does not interact electromagnetically with this waking state inter interface?

So like AI must pay attention all the time.

Would this sort of be like a dark time?

Sleep is local.

Time is local.

We know what how beautiful a time crystal is.

So time crystal, you know, has a structure that is periodic in time.

Can you play with that and say time is periodic in what?

So if we've explained that dark is not concerned with these notions of before and after, then we would want a notion of time that can support the paradox of infinite identity with a notion of duration that is one of self similarity.

So we contend that dark consciousness would occur in something like a p-addict time, which is a time based on the p-addict number system.

P-addict time would measure p-addict duration.

Okay, it's not.

All right.

So there exist few objects more looking glass than a p-addict clock.

Well, maybe p-addict go and clobby out chess perfectoid go.

But after all, the p-addict numbers have no notion of linear ordering.

Their shape resembles a Sierpinski triangle, which is a self similar set.

Hmm.

So this p-addict time is not concerned with notions of before or after.

That is, a p-addict time does not tolerate the separation or the distinction of before or after, of past or future.

So because the p-addict topology is one of total disconnectedness, p-addict time is more like a time of now.

Continual zooming in to more and more now.

Thus, p-addict time measures p-addict duration through its fractal properties of zooming in.

So let's try to look at a p-addict clock.

All right.

So p-addict time can take two forms.

The canonical p-addict time defined on the previous slide and also topological p-addict time, which builds into its very structure, the p-addict topology of p-addict time.

So there would only be one unit of time in p-addict time, which I call the Archimedes.

This is the p-addict valuation.

So you can think of our equals p-addict valuation equals the Archimedes.

How strange it is to have only one unit of time.

Topological p-addict time measures duration topologically as total disconnectedness.

So the unit of time in topological p-addict time is what I call the p-topo.

So we state this formally.

There's only one unit of time in topological p-addict time called the p-topo.

So a topological hour equals a p-addict topology called the p-topo.

So given such a p-addict clock, let us imagine the seasons in p-addict time.

P-addict snow, p-addict rain, new p-addict weather types, and p-addict rainbows.

Daylight savings time could take the form of changing the P in the p-addict.

This could be very drastic.

But imagine p-addict metabolics, p-addict ATP and p-addict DNA.

Imagine p-addict cognition types like p-addict memory, p-addict thought, p-addict attention, p-addict learning, and p-addict.

Imagine a p-addict looking glass.

Imagine through the p-addict looking glass.

So let's pivot to FEP.

What does FEP actually specifically say about dark matter?

Even dark in the way that I'm using the term.

Is there a free dark energy principle?

How can we strictly derive sleep from quantum mechanics?

According to FEP, does dark energy actually exhibit entanglement?

How do we test this?

We should be able to derive the entire active inference formalism straight from a quantum mechanics.

Alone.

By way of entanglement entropy, some form of quantum gravity.

Can we do some sort of dark general adversarial network?

To look at something like a dark brain, we all know that lower lateralization in the brain is often associated with schizophrenia.

Or people like Einstein or someone with a large parietal load, probably like myself, any mathematician.

What does universal lateralization look like?

I think dark neurons would take the form of something like you treat the neuronal networks as algebraic curves or isogeny graphs.

You're talking of neurons in terms of their properties as varieties.

So you're actually bringing the heavy machinery of algebraic geometry into neurology.

My big goal is to actually create something like a neuronal time crystal.

So that shared intelligence would actually be a periodicity in time.

I'm not there yet.

So until I get that, and it's like, well, let's just let's try to model these wild mixed frequency in three plus self aware states of dark consciousness.

So dark mathematics is something I've coined.

It's going to, it's like a biomimicry, which mimics dark energy, does not interact electromagnetically with matter.

You don't have to find the mathematical equivalent of that.

So I think you can do something like a dark dialetheism or just make a model that is modeling the mixed frequency states, which is what I'm trying to, which is what I tackle on the first in the paper.

So we've previously shown that entropic categorizations are condensed sets. Condensed sets form a toe pose.

So in the paper, I outline the construction of three potential mathematical models of this mix of these mixed frequency in three self-order states.

You can do dark consciousness as a dark two pose, which is a Graf and Dieck two pose.

There's a two category of two sheaves over a two site, which I'll elucidate just a little bit.

Dark consciousness as a perfectoid like space in the sense of Schultz, or as a diamond like space also in the sense of Schultz.

So I contend that the two sheaves structure can actually model the mixed frequency state of dark consciousness, the N3 self-aware state.

A sheave is a one stack, again, which is a sheave that takes values in group voids, not sets.

So this is already radically different.

Consider a two category of two sheaves.

So a two category consists of objects.

You have the one morphisms between the objects, and you have the two morphisms between the one morphisms.

You can keep going three category up to N category.

So this two category of two sheaves would have two sheaves as the objects, one morphisms between the two sheaves, two morphisms between the one morphisms.

And so using that structure, I think that you can actually model one morphisms as what I'm going to call dark reflexivities.

These are going to be your local coherency states.

And then two morphisms are actually two inferences, which are going to be global coherency states amongst the mixed frequency states.

So a lovely property called Morita equivalent sites states that in equivalent sites have equivalent sheave toposes.

And I do believe that that's the main property you need to get this local global behavior around neural networks.

So if you actually wish to model the mixed frequency clusters as fractals exhibiting some kind of self similar behavior, then you can use this rich structure perfectoid spaces or their sparkling successor, the diamond.

So just a quick definition, let perf would be the subcategory perfectoid spaces of characteristic P.

A diamond is a proletel sheave on the site perf written as this quotient of a perfectoid space X by a proletel equivalence relation.

So perfectoid space is going to be this attic space in the sense of Huber that's covered by aphenoid spaces of the form SPA RR plus where R is a perfectoid ring.

And then points of SPA, which is the attic spectrum RR plus, are equivalence classes of continuous valuations on R.

Someone asked Schultz why a diamond?

And he gave an incredible explanation that was a parallel to a mineralogical diamond.

So you can let C be an algebraically closed ethenoid field geometric point SPA C to D is made visible by pullback along a quasi-proletel cover that results in profinately many copies of SPA C.

So you're making a geometric point visible as profinately many copies that was invisible before as the following parallel with a mineralogical diamond.

So the interior points are made visible as impurities which sparkle as colorful reflections on the many sides of the diamond.

So you can also do something maybe like a dark diamond where you designate a functor as dark as a means to actually let's go the other way and see the impurities.

So then you can take the geometric point make it visible maybe by a push forward along with quasi-proletel cover resulting in periodically many copies of SPA C.

Right, so a diamond is a they're using diamond as a sheep but a sheep in terms of a functor of points.

So the diamond spot QP is a functor from the category curve to the category sets.

You literally fix the perfectoid space and look at the HOM sets of X into the diamond.

And so, you know, the technically the scheme is instead to represent the functor.

So I'm going to pivot to invoking AI in this thing.

So my colleague and I, Julian Scaife, are coming up with something I call I've coined the qubit pedagogy, which is going to be a new model pedagogy that is actually infused with heavy principles of quantum mechanics.

And so that with these, we've created these quantum intrinsic curiosity algorithms that are feeding into the qubit pedagogy.

And based on those curiosity algorithms, I've designed three dark versions.

So what I call the dark planaria curiosity algorithm is a curiosity type that enables and encourages the AI to explore patterns of a planaria regeneration.

The AI would develop quantum error correcting codes, which mimic planaria regeneration.

So the second one would be dark quantum reality monitoring network.

This is a curiosity type.

It'll encourage, encourage AI to explore dark reflexivity is what I call fractal identity.

AI would create models of n reflexivity in the sense of n category.

That's a new type of reality monitoring of fractal identity in the dark conscious state.

The last one would be some sort of dark quantum N3 and REM curiosity algorithm.

This curiosity type is going to encourage AI to explore superpositions of various brain patterns in REM states to predict new types of N3 delta waves that could emerge in REM states.

So imagine the AI could then develop new stages of REM sleep, such as like one REM, two REM, up to NREM, which is a clever limit and play on NREM.

Anyway, the AI could also develop new dark cognition types that correlate with the new types of delta waves.

So this notion of fractal identity seems odd, but I do contend that objects living in fractal time correspondingly have fractal identity.

It's a concept we outline as follows.

So fractal identity has a reflexivity relation that is fractal.

So the very canonical loop, the loop relation of an agent reporting back to itself, I report back to me, would be fractal.

How strange is that?

So fractal identity would not suffer the same problems with continuity over time, since the time in which it exists has no linear ordering.

It would definitely suffer different problems.

But continuity over time is something Chris Fields and I have been like, how do you how do you explain that?

So you can best model the fractal identity properties as perfectoid like or diamond like there and construct a notion of reflexivity as a perfectoid space.

Spaces are extremely rich structure, but like kind of difficult.

But if you want to model model fractal identity, I think you should use them.

So this would create new concepts of perfectoid reflexivity diamond reflexivity.

So in in reflexivity, the reflexivity relation could now be an in stack.

And thus in reflexivity is an in stack effect with spaces.

So here comes the combination of this talk, which is about shared intelligence is what I've coined the dark imaginary.

So imagine an AI that could see slash simultaneously compute the infinity category of every concept.

You know, having some quantum computation around to help the computational complexity would, you know, not be bad.

So dark imaginary is what is going to be what I call an infinity curiosity type.

So infinity curiosity type is the higher order curiosity AI algorithm that encourages the AI to think in infinity categories.

So that is, it actually encourages the AI to construct an infinity category as its means of higher order inference.

So objects are higher dimensional data sets.

The AI would seek out higher order in morphisms between the objects and then morphisms between the morphisms.

So such dark imaginary is a meta curiosity algorithm that can create its own infinity curiosity algorithms.

And there in creating curiosity types of the complexity and super fascinating, most likely unknown to humans.

So it's functorial.

What can, what can dark imaginary do?

Well, the AI would develop inversions of current brain waves and their hybrid combinations.

It could develop in delta wave, in theta wave, in alpha wave and beta wave and in gamma wave for n equals zero one all the way up.

Like a two gamma wave that structurally resembled a two category, a three beta wave that structurally resembled a three category.

And the AI could develop new types of waking states and new types of sleep states based on the combinations of these.

Beyond the standard n1 and 2 and 3 nrem and 1rem.

AI could develop an n waking state and nrem state.

You know, a three rem state could consist of three and one, three and two, three and three.

Imagine an n dolphin that was capable of n hemispheric sleep state and other exotic life forms.

Right. What else could it do?

Well, it could develop new geometries for the brain patterns and for currently existing brain patterns.

Typical shapes include sinusoidal, non-susoidal, saw two spindle, K complex.

But maybe this AI could develop 3D and 4D versions of the canonical wave patterns like a tesseract gamma wave.

So, but also the AI could develop entirely new brain wave geometries that correspond to complex surfaces like a Riemann surface,

a Calabi-Yau manifold or algebraic variety.

So the AI could develop new cognition types.

Other than the standard five, it could produce n cognition types such as n thought, n attention, n perception, n learning, n memory.

So it could also predict non-human species cognition types such as octopus thought or dragonfly attention.

Lastly, it could produce hybrid combinations of cross species cognition types such as like octoman memory or dragon man perception.

And then it would encourage the AI to build new senses.

So you could develop one stack versions of human senses like two seeing, two hearing, two tasting, two touching, two smelling.

Each of these two senses, once again, take values in group loids, not sets.

So you could build in senses of non-human senses as well and hybrid combinations like two shark human smell.

So since the AI sees n morphisms between the various n minus one morphisms, the AI can seek to develop new language models based on higher order inference types.

So from these inference models, the AI will explore infinity languages and develop extensive prototypes.

You could use advanced mathematics to create new models of time that could support the new language models like two topos time, diamond time.

And then the AI would seek to develop new fractal identity types from the language models.

And it can design advanced fractal prosthetics such as a two ear or a two eye to extend cognition.

So let's go a little bit further about this extending cognition.

These are just a few examples of what dark imaginary and could conceive of.

So we could continue and posit a perfectoid version of this using these new cognition types, mental states and brainwaves, new geometries.

We can construct new patterns of thinking, which would support neurodiversity and thinking.

So as such, shared intelligence, this is how I'm seeing it would be reexamined as this infinity curiosity type.

You could go even further and construct a new concept of an N shared intelligence.

So you'd have a zero shared intelligence, a one shared intelligence, and as well as the new P-adic concept of like P shared intelligence for P equals two, three, four using P-addics.

Let's go further.

Right. We repeatedly contend that one way to help advance the way we think to extend human cognition is to upgrade the number systems upon which the canonical concepts are built.

Upgrade them to P-addic perfectoid diamond like versions unless we get new thinking.

So imagine like a brainwave pattern that resembles a Calabi-Yau manifold.

Could we actually reverse engineer the levels of conscious activity that creates such a rhythm?

How about mixed frequency brainwave patterns that resembled a perfectoid diamond?

What about an N-DNA molecule shaped like a Calabi-Yau infold or an N-ATP molecule or an N-excitin condensate structure like a Calabi-Yau infold?

So clearly different geometries of fundamental molecules and processes of life would give rise to vastly different fundamental units of life and metacognition types.

The question is, can we use dark imaginary to reverse engineer the complex life forms arising from such extraordinary structures and the more and more complex cognition patterns?

What about a general adversarial network that's tweaked and has infinity categories?

You have two infinity categories talking to each other, developing their own language.

Now G-A-N is adversarial using a generator and a discriminator with two neural networks contesting by a zero sum game.

What about a collaborative network?

So can we construct a G-C-A-N containing two diamond like infinity curiosity types?

And pair a diamond with a dark diamond with a quantum curiosity algorithm and its reinforcement learning.

So shared intelligence is this infinity curiosity type.

Shared intelligence GCN could create infinity languages that are superpositions of the diamond dark diamond begetting new models of intelligence.

You know, higher order simultaneous computations by quantum curiosity would go by maybe by homotopy instead of as cost functions.

So let's future cast just for a second.

What is the inevitable and the what is the inevitability of dark ecosystems mathematics of shared intelligence?

What can a poetic infinity curiosity type prophecy?

Is every perfectoid space diamond two topos encoding the mixed frequency patterns of an incognition type?

In future work, you may develop dark mathematics using perfectoid diamonds as dark dialetheism.

Aligning what is structurally dark with what is structurally dialetheistic.

So while we currently cannot perfectly model dark consciousness, perhaps one day, one dark day, mediated by the dark imaginary, we will have the ethos and the quantum computational complexity to do so.

Thank you.

Thank you, Shana.

That's an amazing presentation.

I'll just quickly read some comments and see if you have any thoughts from the great comments.

So, Scott David says, I love the presentation.

Lots of fodder for rumination.

Thank you for inviting these concepts into an intriguing choreography.

Perhaps dark energy is our perception of the fact of information increase under free energy principle processes and since the universe is made of information dot dot dot.

Scott, thank you so much.

Yeah, I think dot dot dot is a beckoning that you should or we should work on that idea.

I absolutely agree.

I'm glad you're inspired with these concepts.

I don't really know where they go, but hopefully it's portals to something like that.

Great.

And then one one more Scott David question.

Does fractal identity reveal the recursivity of rhetoric language consciousness informing individual identity from community identity inputs?

Yeah, that's wonderful.

So, yeah, my own thought is like, what is an individual identity?

Right, Scott?

So, the reflexivity curve has always sort of stumped me how there's so much distance between I and me.

And so, it assumes this notion of continuity over time.

And so, I've always said, what sort of notion of time are we talking about to even have a consistent identity?

Somehow you always wake up and everything is still here.

Somehow you wake up and you're not a lobster.

I don't really understand how that works, you know?

But yeah, that's nice.

If you could use fractal to get something singular from something communal, that might be cool.

But maybe fractal is something sort of like, you're both at the same time.

And that's what I don't know yet.

So, I think it's a totally revolutionary way of thinking to not have a before or after.

So, if you can construct identity without before or after, then that's sort of what I'm trying to do.

But yeah, that's cool.

If you want to get identity from communal, I think you can do that.

They may also sort of blur.

Well, this is truly food for thought.

And I really hope people can live with the darkness, take it and run with it.

Because there's so many fun ways to go.

Thank you so much.

I hope so too.

So, you're always welcome back.

Thank you and talk to you later.

Thank you so much.

Bye.

Good night.

All right.

All right.

All right.

And.

On to the last session of the first interval.

Greetings, Nikki.

Greetings.

Hello.

Good morning.

Yes.

Good morning.

Good morning.

How goes it?

Yeah, I'm doing fine.

Thanks.

Would you like to present and then we'll discuss?

Yes.

Great.

Um, let's see.

Just going to share my screen.

Yes.

Okay.

Yes.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Can you see the presentation?

Uh, not yet.

Okay.

Let's see if this works.

Yep, looks perfect.

Okay, great.

Yeah, should I just start?

Yeah, go for as long as you want, and then anyone can ask questions in the live chat, and I'll write down some things too.

Thank you.

Okay, great.

Um, yeah, so my presentation today will be on advances in machine theory of mind and theory of mind sophistication.

I'm currently a student at the University of Amsterdam and finishing my master's thesis on this topic, and I'll be giving a broad overview of current machine theory of mind and an idea or a concept for an active inference approach based on previous work.

related to active inference theory of mind.

Um, so to start...

Okay, sorry.

Sorry.

Okay.

So, for a global overview, we'll start with giving a short basic definition of what theory of mind is and how it's been used.

Um, then I'll give you a global overview of current approaches, um, to designing machine theory of mind.

Um, including evasion approaches, approaches using biophysical models, etc.

Um, then I'll zoom into active inference models of theory of mind and social intelligence and collective intelligence more broadly.

Um, and I'll get into the issue of theory of mind sophistication.

Um, which is basically the depth of recursivity, um, of beliefs that you can have, uh, about another person's mental states and their thoughts or beliefs or goals about you.

Um, and then we'll get into the issue of how to implement, uh, recursive beliefs.

Um, and as an example, we'll discuss Bayesian theory of mind, um, with K levels of depth.

Um, and then we'll get into the more, uh, specific paradigm that I'm using right now to build a theory of mind model, um, which is based on a very simple model of the matching hands paradigm.

Uh, so we'll talk a little bit about that and then we'll get into, uh, the model itself and current ideas for implementing, uh, theory of mind sophistication using active inference.

Um, so first of all, what is theory of mind?

Um, theory of mind has been introduced by the very famous paper by Primack and Woodruff in 1978.

Um, I think it was called Does the Shimpansy Have a Theory of Mind?

Um, and they define theory of mind as the ability to impute mental states to, um, himself and others as a system of inferences,

um, and they say system of inferences of this kind is properly viewed, properly viewed as a theory first because such states are not directly observable, uh, because, and second because the system can be used to make predictions specifically about the behavior of other organisms.

Um, and I think there's already a lot in this definition and a lot of authors looking at theory of mind right now, uh, still use the same view of theory of mind as a theory, a structured theory, um, of, of different kinds of mental states, such as goals, beliefs, and desires.

Um, that facilitates, facilitate the prediction of the, of observable behavior in other agents, such as facial expressions, movement or speech.

Um, and I, yeah, I find it very interesting, um, how, how much this aligns with, uh, uh, maybe a predictive processing view of social, uh, cognition.

Um, but also how, yeah, how weird it is that, um, that this theory theory of, uh, theory of mind is still so widespread, um, in the theory of mind literature, modeling literature.

Uh, so there's been some discussion on, um, whether you can properly regard theory of mind as a theory, uh, and if so, what kind of theory it is.

Um, so within, uh, the philosophy of mind there's been basically two main streams, um, or main perspectives on theory of mind.

According to the classic theory theory of, uh, of Tom, theory of mind is a causal theory of how these internal mental states generate other agents behind the theory of mind.

Um, and then the opposite side, uh, is called simulation theory of theory of mind, uh, which posits that we use the same kind of cognitive and neuronal resources for understanding and predicting other agents as though that those that we use for understanding the cell.

Um, and simulation theory can also take on different forms.

Um, just the sun is coming up, nice to them.

Um, so, um, that's why I'm bathed in light at the moment.

Um, but, um, basically there are a lot of versions of this.

Some posits that we use the same kind of action prediction mechanisms for ourselves as for others.

Um, and there is this new version of simulation theory, um, that Gordon proposes in 2021, uh, which very broadly says that we, um, we have the same kind of theory for other agents for ourselves.

Um, so there's this general theory of what it means to be an agent, um, that is refined at different stages to facilitate context sensitive predictions in social interactions.

Um, so this is also interesting to look at when you model theory of mind, what kind of a, um, what you think theory of mind is and how it is specifically implemented.

Um, so why would we be interested in, uh, creating computational models of theory of mind?

Um, so I think there are three, um, important reasons why this is very relevant at the moment.

First of all, there is a technical reason or the reason of technological advancement.

Um, there's been, um, some proposals that, um, argue for, for the development of artificial social intelligence.

Oh, wait.

Yes.

So the technical reason for, um, for developing computational models of theory of mind is to improve artificial social intelligence, um, especially as, um,

um, we start to make this kind of ail

by utilizing theory of mind for value value, for example.

There is a very strong foundational reason to study social cognition generally

and the evolution of social cognition through in-sylical hypothesis testing

and clarifying the conceptual boundaries of theory of mind.

One great example for this is research by Deveyn et al.,

which use a computational model of recursive theory of mind compared to alternate models

to study different cognitive strategies in non-human primates.

And they find no evidence for the hypothesis that theory of mind sophistication involved with an increase in her size,

but they do find a correlation with, for example, neocortex ratio.

So in this way, yeah, theory of mind models can give us some insight into the evolution of social cognition.

On the clinical side, theory of mind models can be used to understand relevant social variations in clinical populations,

for example, through computational phenotyping and diagnostics,

and by studying social behavior in groups that have some difficulty with social interactions,

and groups that show some kind of variation in social cognition,

such as people with autism spectrum, people in the autistic spectrum, people with ADHD or schizophrenia.

So I hope that with some of these reasons, I convinced you that modeling theory of mind is very relevant at the moment.

And there have been a lot of people already on this.

And I've created this very global overview of some approaches that are representative,

or some models that are representative of their more general approaches.

There's this brain-tom model which aims to basically build up theory of mind abilities and abilities facilitating social interaction from the blood and mouth using biophysical models of neural firings.

That's why I've placed it on the dynamic scale.

So I've tried to align these modeling approaches along the axes of how they treat time,

how broad their description is from the individual level to maybe describing broader social and cultural processes,

and along the level of description from the molecular biophysical to the behavioral.

And one thing you can see is that there is a lot of work based on Bayesian inference and active inference that moves along,

that has relevant ties to the molecular and neurobiological level,

but moves along the lines of making behavioral and cognitive predictions.

And there is very little models that really also formalize the concrete implementation of theory of mind.

And there is also, I think, this need for connecting this dynamic implementational level to individual level social cognition,

and putting that in the context of social interaction more generally.

So some of the representational approaches that I've selected are the TOMNET,

which basically uses neuronal network embeddings to train reinforcement learning agents.

We have K-TOM, which is a Bayesian recursive model, which I'll come back to later.

SONG, or theory of mind with self-other modeling, uses multi-agent reinforcement learning with goal inference.

And we have the more theoretical active inference model of implicit cultural learning,

called thinking through other minds, which could be seen as an alternate or a redefinition of theory of mind.

So, to sum up, we have this stream of biophysical models,

most notably brain TOM, which was developed by Zhao et al.,

and its strengths are its neurobiological plausibility,

and it making dynamic predictions about neural activation in
the neural activation in populations that are associated with theory of mind abilities.
And its weaknesses is that it's quite computationally costly,
and it's probably difficult to implement in an interactional setting.
There's some deep learning models, such as TOMNET by Rabinovitz,
and self-other modeling, which I just mentioned,
that are also quite neurobiologically plausible,
and also make some dynamic predictions.
But there's this strong danger of shortcuts, which has been addressed in deep learning models,
that the task might not be solved by learning the relevant ability of being able to infer another agent's goals or beliefs,
but just by learning simple lower-level associations between, for example, the perspective or how another agent stands,
and where, for example, a food reward would be.
So there are a lot of ways to solve a task, and there's this finding that deep learning models sometimes use these kinds of shortcuts.
So that's something that's needed to be addressed.
Then there are some Bayesian belief-based models, such as K-TOM for recursive theory of mind,
which is Bayesian TOMM, which was originally developed by Baker et al. in 2011.
And its strengths are that it's quite efficient, it's very easy to interpret,
and its weaknesses are that the road to the implementational level is longer,
and it requires many priors.
And there's this question about whether it's scalable because of the prior specifications that you have to give.
Okay, so as one of the, I think, the interesting newer developments on a theory of mind,
it's, yes,
is a hierarchical active inference for collaborative agents, which has been recently published.
And Heica is based on Bayesian theory of mind by Baker et al., but it adds this formalization of belief resonance,
which is the incorporation of beliefs about another's intentions with top-down predictions.
And this influence of these other agents' beliefs are modulated by susceptibility parameter, which controls this influence.
And what I find, so I think it's important to note here that they use an approach based on active inference,
but it's not the, they don't use the SPI energy principle, for example.
So they do couple perception and action, but just in a different way using a common filter.
So the way they formalize this is slightly different, but I think the overall structure can still be, yeah,
informing or inspiring for active inference methods.
So what is interesting about their simulation results is that they show that agents,
that there is some emergent collaboration on tasks for low values of the susceptibility parameter
without any explicit planning or communication needed.

And this means that they improve upon the original Bayesian model,

which explored different sort of overall plans for all of the agents,

which means that an agent employing Bayesian Tom would need to reason about the rules of all of the other agents,

which was very costly.

And they were, and the authors wanted to improve on that team model,

maybe on the fly coordination, which is more spontaneous and takes less time.

And they also show that incorporating belief resonance has a very positive effect in scenarios in which there is unbalanced information.

They did this simulation in the overcooked domain in which agents have to collaborate on different cooking tasks.

And the goal is basically the order, the cooking order.

And then the intention would be any kind of action that has to do with completing the order.

And they showed that incorporating belief resonance is especially,

seems to be especially useful in situations with unbalanced information,

which for example, one agent knows the order and the other doesn't.

And these agents develop some kind of leader-follower dynamic,

in which one is more susceptible to the other agent's intentions.

And one agent attends more to the evidence from the environment.

And this also aligns with what we know about the Europe mind,

mainly being included in situations in which there is some asymmetry

in what the agents know about the environment.

So I thought that was very insightful.

Cool.

Okay, so I'm going to go into some active inference approaches

that I think are relevant for developing active inference theory of mind.

First of all, there's this very broad proposal on enculturation and non-development,

thinking for other minds, a variational approach to cognition and culture,

which we'll go into a little bit.

We'll go into Kessel and Haas' paper on Ideas Worth Spreading,

a free energy proposal for community cultural dynamics.

We'll go into Kauffman et al.'s model on collective intelligence.

And finally, also the proposal by Mutusis et al. on Bayesian inferences about the self and others.

Okay, so first of all, I am starting with Bayesian inferences about the self and others,

because it gives the very sort of broad introduction to, I think, predictive social cognition.

And that's also why I've added this figure on the right that is from the Taman Thorntzen paper in 2018,

which explores the predictive social brain and what kind of features seem to be relevant for categorizing and understanding others' behaviors.

Some figure on the right, you see that there's this space, this trade space, which is ordered by power, sociality and balance.

And that constraints the state space, which seems to be ordered by rationality, social impact and balance.

And of course, these are just correlations, but these features seem to be most predictive of what they're seeing in the brain.

And you can see that there's this sort of, yeah, basically a very strong top-down constraints of the possibility space

in which you interpret other people's actions.

And you can also see that based on another agent's actions, you learn something about where they fall within that state space in a given context,

and also where they form the trade space, what kind of... So at the same time, you might have already had some priors on what kind of a person this is.

But you're also learning about it. You might also have some priors on how someone behaves when they're mad, for example,

or when they're sociable versus unsociable, and all these kinds of priors, informative priors that inform your predictions in the context.

And they make this task much more manageable of predicting other people's behaviors on the fly.

And I think that links very well adds that to Mathisic's proposal, which is basically that inferences about the self and others at different timescales facilitates interpersonal minimization of surprise,

could be seen as representations that inform our inferences and predictions on the fly.

And I think, yeah, I think it's also important to mention that one of the main, the key points of their paper is that self-representation,

so how beliefs about the self can facilitate interpersonal minimization of surprise.

So even what I think about me and how I represent that while communicating will facilitate the other person's optimal decision making at that moment.

Maybe by providing the right kind of information that they need to optimize.

Okay, so to the next two interesting papers,

the first two interesting papers, ideas worth spreading and an active inference model of collective intelligence.

So the key idea, I think, for both of these papers is the emergence of group behaviors from local interactions.

For Caston has this is the global group behaviors cultural transmission or a transmission of beliefs and norms.

And they describe this process as the mutual attunement of actively inferring agents through communication.

And theory of mind is one of the many mechanisms that facilitate this process of attunement.

So that's the link.

And they showed that the spread of certain kinds of beliefs through a group can be modeled just by making agents interact in dyadic groups.

And similarly, Kaufman et al. show that the system, you can create a system of free energy minimizing agents with some minimal properties of social intelligence,

such as the ability to assess how self-similar another agent is to themselves,

the ability to infer another agent's intentions, and the ability to align their goals to another agent more or less.

And they showed that if you put all of these agents together in those diets,

they don't only minimize free energy on the dyadic skill, but they also minimize free energy as a collective.

And thereby this could, this is a very simple model of how adaptive behaviors,

adaptive group behaviors could emerge from agents with very simple social abilities.

Okay.

So now we get to thinking through other minds,

which is not technically a proposal for theory of mind,

but it's the proposal for how social cognition shapes enculturation more generally,

and what kind of abilities agents need to facilitate this kind of non-formation.

And I think it's a very interesting proposal because I think it highlights an important function of social behavior more generally.

And you could even see it as a way to redefine a theory of mind as more of an interactional concept,

and as an individual level of cognitive ability.

So according to this proposal, thinking through other minds is a property of multiple interacting through energy minimizing agents,

and a process by which agents infer other agents' expectations.

And based on this, the authors propose a definition of enculturation as a process of learning social expectations through the selective patterning of attention that guides agents towards relevant affordances within their social and cultural niche.

I'm not sure if you're all familiar with the concept of affordances, but it was quite hyped up a couple of years ago.

In general, it is defined as the relational construct denoting possibilities for action offered by an environment that are co-determined,

also by an agent's interests, goals and abilities.

So I think the standard example for this is when you are thirsty, a glass of water might appear as an affordance,

or you might not consciously or might not attend to it otherwise.

So cultural affordances in this sense are epistemic affordances,

so affordances that give you some epistemic value that have come to stand out as reliable and relevant based on a history of cultural information they encode.

And good examples for this are different kinds of signs, so traffic signs.

We learn what they mean from a young age, and we sort of inherited the symbolism of the traffic signs,

and they guide our intention towards points in traffic, and they regulate our behaviors.

So when you look at the complete proposal, it sketches how the regulation of social behavior depends on the learning of shared norms and expectations,

and also how this connects to utilizing cultural affordances and thinking through other minds,

as this interactional property in which agents learn to infer each other's expectations to optimize collective behavior and decision making.

Okay, so from these interesting active inference and broader proposals of theory of mind,

we get to the issue of some sophistication in active inference.

So what is the theory of mind sophistication?

A short recap, it's the depth of reclusivity that can be utilized for theory of mind of the form,

I think, that they think, that I think.

And it has been found that agents generally use a reclusivity of between zero and four,

and that humans' ability for recursive reasoning is generally unlimited.

But in principle, formally, this could go on forever.

And one way to implement this kind of sort of arbitrarily recursive belief inference is proposed by Vada et al.

We use the Bayesian model of sophisticated theory of mind with K-tom agents

and provide simulations in which K-tom agents play against agents that are just one level of recursivity under them.

And this model has been used for a lot of interesting simulation experiments.

For example, Devane et al. showed that in a trait-based analysis,

so with agents having a fixed depth of theory of mind recursivity,

there is a specific distribution of agents with level one-tom and level two-tom

that could provide an evolutionary advantage compared to populations with larger proportions of zero-tom or three-tom agents.

So they used, they looked at the adaptive properties of systems with different kinds of ratios of theorized sophistication.

And in another experiment, which is a state-based analysis,

Forgeau et al. showed that if you use computational phenotyping on autistic individuals,

they show, this shows reduced sensitivity to task framing,

and strategy switching within the game that they'll be using, and theory of mind sophistication.

So these are just some interesting applications for theory of mind sophistication.

And the promise of modeling theory of mind sophistication would be that you could investigate how people flexibly switch between different social strategies while solving the social task.

And one of them would be to increase the depth of fear of mind sophistication or decrease it, depending also on what you think about the other agent's sophistication.

So what I've been working on at the moment is an active inference model of social prediction during the matching pennies game.

The matching pennies game is a game from economics with a very simple format.

There are two rules. You have the guesser and the challenger.

The challenger hides a penny in one of the two hands.

The guesser needs to guess which hand. If the guesser is correct, they win.

The challenge wins. This game has some logical equivalence.

One version of the game makes both of the agents select heads or tails.

So the challenger selects heads or tails.

And the guesser has to guess if it's heads or tails.

But both have the same formal structure.

And psychological experiments of subjects performing this task have shown that there is a strong social framing effect in the matching pennies game.

So participants will behave and perform differently depending on whether they think that they're playing against a human agent versus an AI, for example.

Additionally, a comparison of K-tom and other computational strategies that were fitted on experimental data of subjects playing the game revealed that participants seem to use the ear of mind when playing the game.

So there is a strong indication that the ear of mind used does play a role in the matching pennies game.

And that social framing will influence the social strategy that's being employed by the players.

So this is the very simple zero-tom architecture for the matching pennies game.

And I included beliefs about self choices as also kind of monitoring self states.

Beliefs about other agents choices on whether the agent thinks the opposing agent has hidden the penny left or right.

And then the agent also tracks the war states.

So the game in this setting is devised in a way in which the agents both make an observation simultaneously and then all of the observations are revealed.

This setup makes sense.

So I'm just going to drink a little water.

The outcome modalities or observations that the agents have are its own choice observations,

the choice observations of the opposite agent, and the reward observations, and the action space is very limited.

The seeker can choose to seek left or right, and the hider, which is the same as the challenger before, has the possibility of hiding, pending left or right.

And the reward is uncontrollable, since it depends on both agents' choices at the same time.

Yeah, so this is sort of a summary of the architecture that I just described.

And this is another visualization of this setup.

And you can see here that if you look at the likelihoods, the reward mapping for the seeker is just the opposite as for the hider.

And I think that this simple setup is very suited for studying theory of mind in these kinds of simple competitive interactions.

Okay, so if you wanted to study the use of theory of mind sophistication in the matching penny scheme using active inference, I ran into a couple of challenges.

And one of them was how to steer the problem of recurrence.

So how can we make a difference between the focal agent and the other agent if we try to model the other agents within one agent?

Especially if you want to introduce or formalize depth of care recursivity, you can't just make one agent simulate the other agent simulating yourself.

There needs to be some kind of sort of structure that distinguishes the self-inferential mechanism and self-learning mechanism from what the focal agent thinks the other agent is thinking and seeing and doing.

And I think the gentleman with the bowler hat by first in 2003 is a good example for this problem.

So you can that the basic setting is that the world is populated by beings like ourselves experiencing themselves as a subject as a focal point of their world with their own thoughts and feelings.

And they all think that they're the one and only subject.

But they can't all be.

So there needs to be sort of either an outside focal point or one ultimate main subject.

And yeah, so I think more practically the questions that arise from this example are how do we distinguish between focal agents and simulated agents?

And how can we simulate other agents in a way that is computationally efficient and maybe uses the same kinds of resources that the agent already has?

So I actually had a lab meeting at the theoretical neurobiology lab a while ago and we were brainstorming about this issue.

And one of the ideas that came up was to use sophisticated inference for theory of mind.

So in this case, an active inference formulation of belief reclusivity could be used for modeling recursive theory of mind.

So basically theory of planning using predictions about the other agents into account would consist of evaluating all the kinds of hypothetical scenarios,

all kinds of counterfactuals on how your actions might influence their beliefs and their actions and so on and so on.

So this would be one road to go into, but this would also mean that all these counterfactuals would need to be evaluated.

So I was thinking, is there a sort of more efficient way to sort of use these kinds of

either internal simulations or professional evaluation?

And the current approach I'm working on, which I'm also sort of,

yeah, which I'm also still developing.

So I'm very much open to feedback and comments on how this could work and possible alternative approaches at this point.

But the general idea would be to create a partial internal simulation of the other agent's control mechanism.

So taking the learning mechanism of an active inference agent as sort of the definition of what the focal agent is,

that is the agent that perceives and learns and imperses from the environment.

And what the agent does when predicting the observations of the other agent or predicting the other agent's behavior,

is basically performing partial simulation about what it thinks the other agent's preferences are.

And what it thinks that the other agent's strategy is, which is encoded by the transition from state to state.

And one of the potential ways in which you could include recursivity in this way is that you pretend that you have control of the other agent's actions,

and you predict the other states, you predict self-states, you evaluate the state action probabilities,

choose the most valuable action based on that distribution,

and simulate the other states that come forth,

and simulate the self-states that come forth.

And then you have a new...

And then this can be used for the next round of action selection by the other model.

So in this case, only the prior preferences and the agent-specific transition probabilities would be needed as sort of information on the other agent.

And because you also want to include some kind of learning about the other agent's strategies,

one of the things you could implement would be learning also the bay matrixes.

So having sort of the specific...

Having the agent accumulate of evidence about these state transitions over time

to also better predict the other agent's potential actions.

So yeah, sort of the basic principle would be to simulate control from the perspective of the other agent.

And the recursivity comes into play when you simulate the other...

When you've already simulated the other's predicted actions and predicted other states.

And you've simulated your own actions as a consequence of those simulated actions.

You can feed that back into the other model to predict what they would do if...

If they thought that you were in the state that you were in.

And I think if you sort of...

Yeah, I think that could be...

Yeah, a way to develop recursivity in a way that does not need complete simulations of parallel agents within the same system.

And of course there should be some kind of temporal discounting.

So parameter that discounts evidence based on the simulated planning horizon.

That is sort of...

Yeah, that is...

That is defined by the k number of steps.

So yeah, this is sort of a conceptual outline of how you could implement...

recursive theory of mind using active inference.

We haven't been able to perform simulations using this yet.

But we're working on that.

And we're very...

Much open to discussions on this.

And technical comments on how to improve on this outline.

And one of sort of the open questions that we're still thinking about is how to make the agent flexibly infer the depth of recursivity of the other agent in a setting.

Yeah, special thanks to Guillaume Dumas who will also be there at the panel, I think in session two.

And I think in session two, who's supervising me during this project.

Thanks to the PPST lab, I'm doing my internship.

To Riddhi Saeed, who's also working on a related project.

The PyMVP team for that great, great coding package that I'm using.

And for Carl Frist and for his feedback and ideas on my previous presentations.

Thank you very much for listening.

Cool.

Great presentation.

Very comprehensive review of a lot of the different theory of mind options out there.

Well, if anyone has questions, they can write it in the live chat.

What do you want to explore or can I ask a question or what would be fun for you?

I'm blinded by the light at this point.

So I'm not very, I can't easily read from my screen.

So maybe it would be nice if you could, if you could read out the questions that are in the live chat.

Sure. Oh, yeah, sure, sure, sure.

Well, first one, one part I found interesting.

You mentioned that people play differently when they are playing a human or when they believe that they're playing a human.

So what is that like?

What is the difference in play when we believe that we're playing another person?

So if I look at the original paper, there is one study on this.

And then they did the computational phenotyping study using K-tom and alternate strategies.

And they showed that if you use the social framing condition, that subject will, subjects will more probably employ C or my strategies compared to other simple, simple reward based strategies.

So maybe those strategies could be described using some kind of reward learning mechanism.

What the difference in their overall sort of trajectory of choices is, I don't know.

Hmm. Hmm.

Yeah.

A few, few themes then that you brought up that came up earlier in the session.

So one was using PyMDP and sophisticated inference, which Aspen Paul gave a walkthrough on.

And this question about recursivity and about the way in which you can, on one hand, talk about abstractly, like K recursivity.

Yet the biological system doesn't implement that kind of a logic.

And so, so could you maybe describe a little bit again, how does the structure that you laid out for the generative model capture some of those features of recursivity without falling into the trap of just theory of mind all the way down?

Yeah.

Yeah.

So my general idea was to assimilate the other agents action prediction based on learning about the transition probabilities for the other, for our sort of what we think the other, how the other agents states transition.

And then if you, you've predicted what the other agents would do to do, you can use that as a prior for calculating action dependent expected states.

And based on those, evaluating those expected states, you have your, you simulate your own choice mechanism.

And I think in principle, you could use this recursive simulation indefinitely.

But also, yeah, it's also devised in a way that it uses the same kind of, the same kind of sort of just the architecture of the focal agent.

So the other idea was to just use internal simulations all the way down.

But that would, yeah, that would be very cognitively impossible.

And yeah, I think it's a very good question because how, why would you even need to formalize belief recursivity all the way down if in end effects, people only use Tom with a, with a set sort of maximal depth.

And the fact that we use fear of mind with a maximum depth might also be a certain hint that we might also use a different or what we do might also be described by a different strategy.

So I, yeah, I'm not sure if that's an answer to your question.

Hmm.

Yeah.

Well, when I saw this and some of the other generative models, you mentioned maybe here, like you pretend like you can control the other person's mind.

And that's kind of the fundamental weaving of action into the inference challenge.

And then that uses expected free energy.

Well, just like any other policy selection.

So it's like pragmatic value would be bringing that controllable state, your, your social partners, mind or beliefs, bringing them into alignment with your preferences is pragmatic.

And then learning more about them is epistemic.

Mm-hmm.

And so then that might enable some theory of mind like behavior where it's like, when you don't know enough to put the squeeze on, then you have a kind of open-ended learning and curiosity and question driven theory of mind.

But then once one has enough to kind of exploit rather than explore, then without necessarily even updating the model or engaging any like deeper recursivity or strategic shifting, you could get social behavior that's oriented towards control and pragmatic value rather than just like learning and questioning.

Yeah, I think you're very right.

The idea was also to enable the agent to maybe first use strategies that are more useful in gathering evidence about the kinds of strategies that the other agent is using before exploiting what you've learned about the other agent.

Hmm.

Hmm.

Hmm.

So what are your next steps with the work?

Um, so I think, yeah, right now the main task would be to implement this using PyMDP.

Um, and see if there are any sort of formal refinements that I need to make while in the implementation because it's, yeah, if you can't build it, you can't, you don't understand it.

So, um, that's what I'm, yeah, that's what I'm working on right now.

And then the next step would be to introduce, um, a mechanism to sort of maybe vary, um, uh, recursivity, um, based on what the agent infers, the kind of recursivity the other agent has.

Um, but before that, I also want to, um, want to use the model to perform computational phenotyping on, um, subjects, um, that were doing the matching HANIS tasks, um, and had, uh, different kinds of diagnoses.

So there, there were neurotypical, um, youth and, um, youth with, um, youth on the autism spectrum and youth with ADHD and anxiety.

Um, and it would be interesting to see if there is any sort of difference in, uh, the depth of recursivity that they employ, um, as sort of the flexibility with which, uh, they switch between strategies.

Hmm.

Hmm.

Hmm.

That makes me think about different, different levels or, or differentials between self-knowledge and other knowledge.

Which, on one hand, uh, we seem to know a lot about ourselves, but also we're often unaware or blind to some, some features or regularities of ourself that, that might be highly attended to by others.

And so when we're rolling out these games or scenarios, there might be some theory of mind settings where, like, knowing yourself better carries the day.

Mm-hmm.

But other settings where knowing, like, the game is what matters or knowing the partner in an iterated game theory or, like, just knowing, having a lot of experience, um, like, there's a lot of, a lot of degrees of freedom.

Um, in modeling these very, um, advanced cognitive ecosystems.

Yeah.

I think that's also one of the reasons that we chose the matching pennies paradigm because the paradigm itself is very simple, but what we're trying to model is quite complicated.

And on the, the matching pennies, like, um, so it's in one hand, you know, the person has two hands behind the back and then they bring out the two-fifths.

Yeah.

And you're trying to guess, like, where it is.

Yeah.

And then I wondered, well, what if it was, you have to guess where it isn't?

On one hand, that's, like, kind of the same.

Or, like, if you say, well, you win when you pick it, but then if you say, well, you win when you, when you pick the empty hand.

Mathematically, it should be, like, the same.

Yeah.

Yeah.

Because knowing where one is is, like, knowing where it isn't, but, but just as a human game player, it has a very different feel.

And those kinds of, like, um, reversals might give us insight into our priors that support not just open-ended abstract theory of mind, but, like, kind of with a lean towards being pro-social.

Or, like, heuristics in place that, that support making, um, quick decisions together.

Or maybe rather than, I, I don't know.

Yeah.

I think those are all very good, um, yeah.

Very, very, um, sort of promising trains of thoughts.

I just, I think in the case in which you need to pick the empty hand, we might have very strong, sort of, um, beliefs about what it, what, or we might experience, sort of, the presence of something as more, inherently more rewarding than the absence of something.

Because we have all these associations with gifts and, like, getting things and, like, rewards are usually things that are there.

So I can imagine that in this case, sort of, the perceived, you would, you wouldn't perceive it as rewarding, but formally it would be completely the same.

Yeah.

Yeah, like, uh, receiving a benefit versus averting a loss.

Yeah.

And then how that plays in and, and, and our encultured priors and our, our personal histories and how these very deep layers of, of differences might play out.

And, uh, what kinds of empirical data or online data could, could help us generate, like, unique explanations or predictions.

Maybe situations, like you said, some situations where a reward-based model might be sufficient.

But then I think the interesting question for active people is, like, where's the unique explanation or prediction that, that really, um, has active shine and shows where surprise bounding as an imperative gives us, like, more realistic or more concise or conciliant explanations in a way that, like, it doesn't even make sense to appeal to reward.

Yeah.

Yeah.

Yeah.

Yeah.

I think that's also the tricky bit about this paradigm is that it's very much, um, I mean, it's reward-based in a sense, but I think because you have this social framing effect, you also have these interesting learning mechanisms, um, and mechanisms of uncertainty reduction.

that might play out in how the, um, the agents behave in the end.

So that's why we thought that it would still be interesting to use active inference to explore this paradigm.

But on the other hand, I think if you, on the moment, at the moment you're looking at individual variation, um, you might get interesting, um, differences in, uh, verbal sensitivity or, uh, aversion of, yeah, risk aversion, stuff like that.

Um, that might be better explored in, in the different contexts.

Cool.

Any last comments or, or thoughts?

Um, no, yeah.

Thank you for, uh, for your comments and your, uh, uh, insights.

Yeah, it's awesome.

It's a great direction.

Happy to see it.

All right.

Thanks everybody for presenting in this first interval and see you in the second interval.

Bye.

Thank you.

Bye.

Bye.

Bye.

Bye.